

DEEPGRID SEMI

Chips made on older processes · Hyderabad, India

PLAIN-LANGUAGE EDITION · v3

Angel / seed round · ₹10 Cr · September 2026

Mature-Node Silicon for India's Defence and Industrial Base

*In plain words: the chips India buys from abroad, why we can make them here —
and now, chip by chip, who buys each one, what it earns, and the rules that guard the money.*

India buys almost every one of its simpler chips from other countries. About \$9 billion of them a year. These are the unexciting parts inside every vehicle, electricity meter, drone and defence box: power chips, small processors, motor drivers, data-line chips, voltage monitors.

Nobody in India makes them. The reason is not skill. It is that designing a chip the normal way costs \$2 to \$5 million before you sell one unit, and these chips sell in small numbers. This document explains how we cut that cost by about ten times, what we have already built, what ₹10 Cr would buy — and, new in this edition, the money one chip at a time, the named anchor customers, and the stop rules we set for ourselves in advance.

\$9 bn

of these chips India buys
from abroad every year

130 nm

we have already made a
real chip using our method

₹2.88 Cr

of real orders today, before
our first chip exists

10×

lower cost per finished chip
than the normal way

WHAT IS IN THIS DOCUMENT

Part One India buys these chips from abroad, and nobody here makes them

Part Two We changed the cost, and we have real chips to prove it

Part Three The method: this process, this loop, this library of blocks

Part Four Ten chips, and what it takes to turn one into a product — now with the money, one chip at a time, and named anchor customers

Part Five Two ways to sell the same chip, what ₹10 Cr buys — and the rule, the scoreboard, the crash case and the stop rules

EVERY CLAIM IS LABELLED

DONE a chip exists, or an order is signed · **DESIGNED** the circuit code works on test hardware
PLANNED a tracked date, not a claim · **ESTIMATED** our own number, with no outside source

There is a word list at the back explaining every technical term used.

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ABOUT THIS VERSION

This is the plain-language edition. It has the same facts, the same numbers and the same warnings as the full technical whitepaper. Only the writing is simpler. Short sentences. Common words. Every technical term is explained the first time it appears, and there is a word list at the end.

PART ONE

India buys almost every one of these chips from abroad Nobody in India makes them. The reason is money, not skill. And money can be changed.

1 What India buys, and why nobody here makes it

Start with one number. Last year India imported \$23.4 billion of chips. That figure is from UN COMTRADE, code HS 8542. It is the only number in this section that comes from outside our company. About 70 % of it came from China and Taiwan. Inside that total, analysts estimate about \$9 billion a year is the kind of chip this document is about. These are chips made on older, larger factory processes — 130 nanometres and above. Power management chips. Small microcontrollers. Motor gate drivers. Data-line transceivers. Voltage supervisors. Electricity-meter front ends. These are not exciting chips. But they are most of the chips in any product. A car has one or two advanced processors and several dozen of these older ones. A smart electricity meter has none of the advanced kind at all. So the import bill is large, and it stays large.

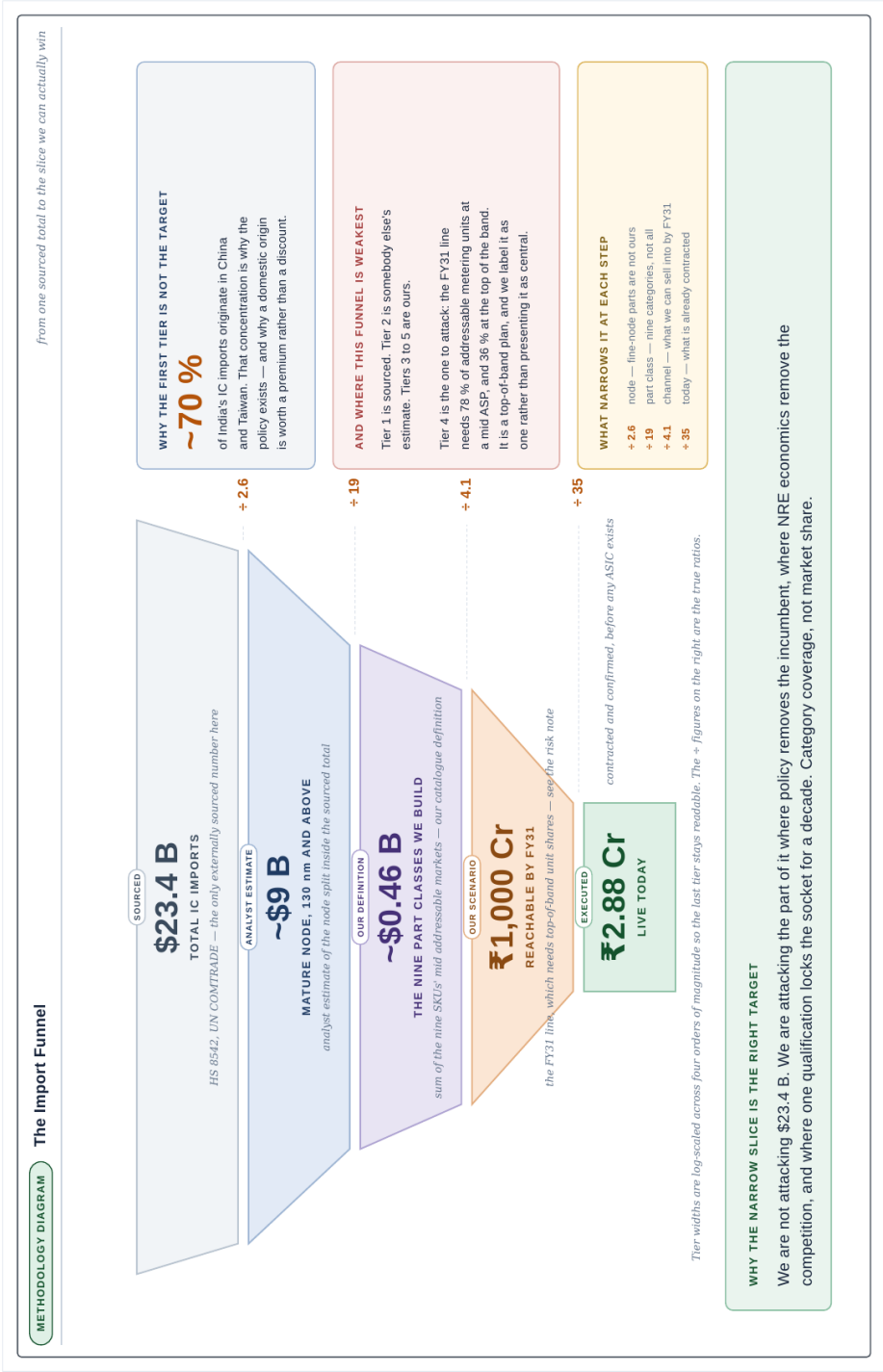


Figure 1. The import funnel. Five levels. Each level has a tag showing where the number came from. Only the top one is from an outside source. The next one is an analyst's estimate. The bottom three are ours. The widths use a log scale so the smallest level is still readable, so the true division is printed at each step.

Figure 1 is a funnel, not a headline. Each step down makes the number smaller and makes it more believable. Walk down it:

- \$23.4 bn to \$9 bn. We cut out chips made on newer, smaller processes. We do not build those. This

divides the number by 2.6.

- \$9 bn to \$0.46 bn. We cut down to the nine kinds of chip we actually build. This is the biggest cut, a

factor of 19.

- \$0.46 bn to ₹1,000 Cr. This is what we think we can sell by FY31.
- ₹1,000 Cr to ₹2.88 Cr. This is what we have already sold. A factor of 35.

We would rather you argue with the bottom number than believe the top one.

1.1 Why has nobody in India filled this gap?

India has plenty of chip designers. So why is the gap still there? The answer is not skill. It is a fixed cost. Designing a chip the normal way costs \$2 million to \$5 million before you sell a single unit. Most of that is not design work. It is:

- physical layout work sent to an outside company: \$1–2 million per chip
- software licences for design tools: \$0.5–1 million every year
- a full set of photographic masks to print the chip: \$0.5–1 million

To pay that back you need volume. Millions of units, over several years, selling to a market that does not change its mind. Now look at what Indian defence and industry actually ask for. A power management chip for one avionics box might sell 10,000 units a year. It needs a specification nobody else uses. It must be screened to a military standard. You cannot build that chip against a \$2–5 million setup cost. Not because it is hard. Because the sums do not work. So the big chip companies do the only sensible thing. They offer a catalogue chip with the wrong specification, or an old chip that is about to be discontinued. The Indian buyer takes it, because there is nothing else. Nobody chose this. It is just what the costs force. And it has been this way for thirty years.

1.2 Then the law changed

Something happened that a pure business analysis would miss. The Indian government stopped preferring Indian chips and started requiring them. Four rules now stack on top of each other:

- DAP-2020 Buy (Indian-IDDM) puts Indian-designed products first. An Indian part beats an import

before anyone compares prices.

- Positive Indigenisation Lists (PILs) ban the import of named items after a stated date. The foreign

supplier is not just at a disadvantage. It is removed from the bidding.

- SRIJAN, a government portal, lists 37,696 imported items that defence companies want replaced. The

buyer publishes this list itself.

- Make-II gives you a guaranteed order if your trials succeed. The buyer risks time. The supplier risks

development money. This combination is unusual, and it matters more than the market size. In a normal market, a cheap new competitor is answered with a price cut. Here, once an item is on a PIL, the foreign supplier cannot be chosen at any price. And once we are on an approved-vendor list, it becomes expensive for the customer to switch away from us. There is one catch in this argument. It is big enough that we will raise it ourselves.

BUY(INDIAN) IS NOT THE SAME AS BUY(INDIAN-IDDM), AND FOR NOW WE ARE THE FIRST ONE The IDDM category depends on how much of the product's value is Indian. It does not depend on where the design was done. Our first chips will be made in a factory in Wisconsin in the USA, and the radar chip in Germany. The design, the circuit code, the verification and the layout are all Indian. The silicon wafer is not. Read strictly, that probably puts us in Buy(Indian) and not Buy(Indian-IDDM), until we are making chips at SCL Mohali in India. Two things follow. First, Buy(Indian) still beats a straight import, and a PIL removes the import whichever category we are in. So the advantage is real. It is just one step lower than the strongest version of this argument. Second, this makes the SCL Mohali factory a business need, not a nice-to-have. That is why we keep it in the plan even though it is the least proven of our three factories.

1.3 Why now, and not five years ago

Nothing above is new information. The import problem has been visible for thirty years. The engineering talent has been here for twenty. Four things changed recently, and all four had to change: Free chip design kits became real. A factory design kit, or PDK, is the file set you need to design a chip for a specific factory. SkyWater released a free one called sky130 in 2020. A German institute, IHP, released one in 2023. Before that, "free PDK" meant a university process with no factory behind it. The free software tools improved too, to the point where they produce files a real factory will accept. RISC-V processors became good enough to ship. RISC-V is a free processor design standard. No licence fee, no royalty, no negotiation. Five years ago choosing it meant accepting a worse processor. The PIL deadlines are now. The dates in §11 fall between 2025 and 2028. A chip designed in 2021 would have been too early. One designed in 2029 will be too late. The smart-meter programme hit its peak years. Our largest commercial volume is a national meter programme, and its peak is in this decade. So the gap exists because of cost. And the law has now created a buyer who cannot fill that gap with imports. The rest of this document asks whether we can change the cost — and whether we have already shown that we can.

PART TWO

We changed the cost, and we have real chips to prove it. Our cost per finished chip is about ten times lower. Not because we are clever. Because four bills are not on our list.

2 Our costs, line by line

Our claim is small and easy to test: Use an older factory process, a free design kit and an automated design flow. Together these cut the cost of a finished, tested, low-volume chip by about ten times. This makes a group of chips profitable that the big companies cannot profitably make. Indian law then keeps the customer with us for a decade.

The important word is cannot. We are not saying we are cleverer than Texas Instruments. We are saying four bills on their list are not on ours. That changes which order sizes make money. Cost item Normal chip company Our way Physical layout \$1–2 M sent outside, per chip done in-house on free kits — staff time only Design tool licences \$0.5–1 M every year free tools — almost nothing Design time 12–18 months per chip design files in under 30 days; chips every 180 days Prototype masks \$0.5–1 M for a full set shared factory run — ₹14.3 L on sky130, ₹34–80 L on IHP Testing and approval rolled into the project cost ₹40–80 L per chip, listed separately Total per chip \$2–5 M ₹0.6–1.2 Cr including approval Three of those numbers are now quoted from the supplier's own price page, and one of them shows we were wrong. A shared factory run on sky130 costs \$14,950. That is ₹14.3 L at today's rate. It returns 100 finished, packaged chips, and gives you about 15 mm² of space on the shared die. We had been saying ₹15–40 L per chip. So for nine of our ten chips we were charging ourselves too much. The tenth one goes the other way. The radar chip uses the IHP factory, which charges by area: €5,110–7,300 per mm². A radar front end of 6–10 mm² therefore costs ₹34–80 L. That is above the figure we claimed. IHP also runs a cheaper option at €2,800 per mm², but you must publish your design publicly under an open licence. A defence radar chip cannot do that, so it pays the full price. One price range for two different factories was wrong at both ends. Figure 6 now shows the real numbers.

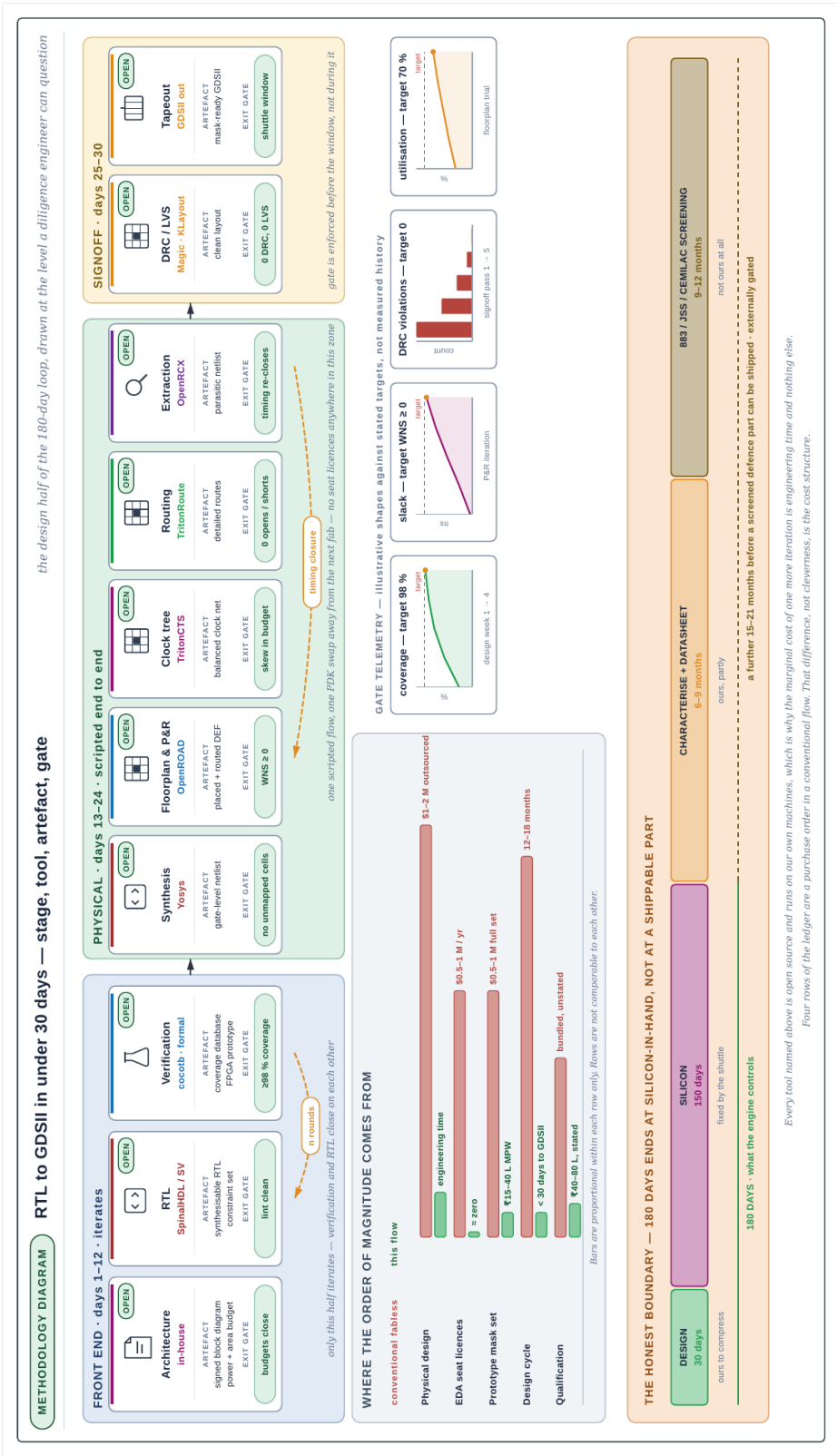


Figure 2. The design half of the loop, at the level of detail an engineer can argue with. Each step names its tool, what it produces, and the test it must pass to move on. The OPEN tag on each card means the tool is free. The table below the flow shows where the ten times saving comes from. The bar at the bottom shows the part of the schedule we do not control.

Every tool in the middle of Figure 2 is free software running on our own computers. Yosys for logic. OpenROAD for layout. TritonCTS for the clock network. TritonRoute for wiring. OpenRCX for extraction. Magic and KLayout for final checks. Four of the rows in the table above are an invoice for a normal chip company. For us they are staff time. So one more design attempt costs us almost nothing. That single fact is what lets a small company build a range of chips instead of one.

2.1 What this does and does not mean

It does not mean we can beat a mass-market chip at ten million units. At that volume the big company has already paid off its setup cost, and it buys wafers cheaper than we do. We would lose. Saying otherwise would be the fastest way to lose your trust. What it means is this. A chip selling 10,000 units a year is a product for us. It is not a product for Texas Instruments. At ₹0.6–1.2 Cr total we can build it, get it approved, and sell it at 60–75 % gross margin. And in this market, the law stops the big company being chosen anyway. This is a low-end entry into a protected market. It is a known strategy with a known way of failing, and §12 deals with that failure directly instead of ignoring it. An argument about cost only matters if the method has actually made a chip. It has.

3 What we have already made

Read this section differently from the rest. Nothing here is a forecast. A 130 nm test chip, made at SkyWater using our own automated flow. This is the key proof. It means our software produced a real design file, a real factory accepted it, and real working silicon came back. Our floating-point maths work at 130 nm is on that chip. Every speed and cost claim in this document rests on it.

DONE **DONE** A design on the TinyTapeout 6 shared chip. One of roughly 500 designs on a single 15×15 mm sky130 die. This is public and anyone can check it. One honest note: we should publish the exact project link, because right now we are asserting this rather than showing it.

DONE ₹2.88 Cr of real revenue, before any chip exists. The Indian Army through MCEME: ₹1.01 Cr under contract, of which ₹23.01 L is delivered and ₹78.39 L is in progress. Infinis Agritech: ₹1.25 Cr for 50 kits, with a 40 % advance already received. Axitech Solar: ₹0.62 Cr for 25 kits. Normally a hardware company raises money, builds the product, then hopes it can sell. We are selling already. And this is board revenue, not chip revenue. All of it is hardware built around FPGAs. FPGAs are chips you can reprogram, useful for proving a design works before making a real chip. So what this money proves is that we can specify, build, sell and deliver a system to an Indian defence or industrial customer. That is not nothing. It is more than most companies have before their first chip. What it does not prove is that we can sell chips. A customer who buys a board is a different customer, at a different price, with a different approval process. Treating ₹2.88 Cr as proof of chip demand would be the easiest way to mislead you. So we are drawing the line ourselves. **DONE** DGS001, our DGFP4 demonstration board, is hardware we sell today.

DESIGNED Circuit code tested on FPGAs. The lockstep microcontroller with fault testing, the motor control datapath, the meter measurement chain, the supervisor sensing chain, and the drone position engine. All exist as circuit code and run on an Artix-7 FPGA at 81.25 MHz. With dates. The IHP factory port. SCL Mohali production. All six main prototypes sent for manufacture by March 2027. Commercial production from September 2027. **PLANNED**

We would rather you find these on this page than in diligence.

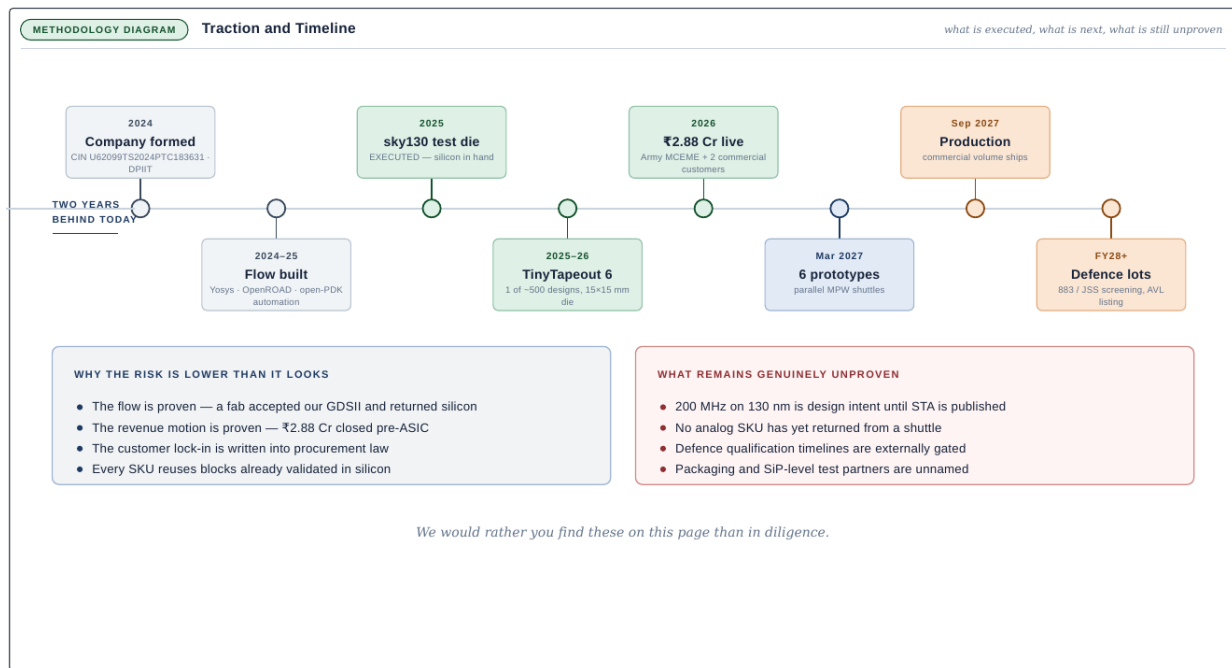


Figure 3. Done, next, and still unproven, in three clearly different colours so nobody reading quickly can confuse them.

3.1 One number people misread, and one that is genuinely open

81.25 MHz is a limit of the FPGA, not of our design. It took eighteen builds to get the Artix-7 prototype working at that speed with the maths engines included. That proves the circuit code is correct and can be built. It says nothing about the real chip. The product is a 130 nm chip running at a fixed 200 MHz. We would rather explain this once, here, than have it read later as a performance claim. Which brings us to the one open item we would raise if we were reading this document. **THE BIGGEST UNPROVEN CLAIM IN THIS DOCUMENT** We say our chips run at 200 MHz. We have not yet published the timing analysis that proves it, and it is not clear which factory kit that analysis was run against. Running that analysis on sky130 and publishing the report is action 2 of seven in §14. It is the cheapest credibility we can buy. We are putting this here, in the evidence section, because that is where a reader would look for it. Cost and real silicon are the two halves of the claim. The next question an engineer asks is whether 130 nm is a real engineering choice or just the cheap option.

PART THREE

The machine: this process, this loop, this library 130 nm is the right process for these chips for reasons of physics, not price. The loop that turns that into a range of products runs every 198 days. And what builds up over time is not the chips. It is the library underneath them.

4 Why 130 nm is the right process, not the cheap one

In chips, people assume newer is better. For this kind of chip that is simply wrong, and the reasons are physical. High voltage works here. 130 nm and 180 nm factory processes offer thick-oxide transistors and LDMOS devices. These handle 5 volts up to 120 volts directly. So a 28 V aircraft power rail, a 120 V motor supply and a 0.9 V digital core can all sit on one chip. On a newer, smaller process the power circuits move back onto the circuit board as separate parts. That is the opposite of what you want. More board space, more parts to buy, more things to fail. This is why four of our ten chips exist at all. Chip 1 has a 5–120 V converter next to a 200 MHz processor. Chip 3 puts an aircraft-standard input stage, an LDMOS power stage and four power rails on one chip. Chip 5 needs a 5 V-tolerant output ring. Chip 8 drives 0–12 V into a display panel across 3,840 outputs. None of these can be made on a newer process at any price. Radiation resistance comes from size. A cosmic ray can flip a stored bit. The bigger the transistor, the more charge is needed to flip it, so bigger transistors are harder to disturb. Protection methods such as DICE latches and triple redundancy cost chip area, and area is the one thing an older process has spare. On chip 3 the power sequencer uses triple redundancy and DICE latches, and this costs a few percent of the chip. On a 28 nm process the same protection would eat into the part of the chip that justified using

28 nm.

HOW FAR THIS CLAIM GOES We only claim design-level protection. We have no radiation dose figure, no latch-up test data and no radiation-hardened approval for these chips today. That distinction matters in a defence conversation, so we state it rather than letting the phrase "radiation tolerant" do vague work. Analogue circuits work better here, not worse. Analogue circuits handle continuous signals rather than ones and zeros. They need accurate resistors, good capacitors and transistors that match each other closely. Older processes have all three. Newer processes trade them away for density, and add leakage and mismatch that the analogue designer then spends area and power correcting. Chip 6 claims $\pm 1\%$ accuracy from $-40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$. That comes

from a temperature-corrected reference circuit and closely matched resistors. For chip 5 the argument reverses completely. A newer process is actively worse for a line driver. A line driver needs a 5 V-tolerant thick output ring with strong static-discharge protection. Newer processes do not have one. This is why interface chips from every major company are still made on older processes today. One place the process is a hard limit. Chip 7 is a 4D radar front end. It cannot be built on ordinary CMOS. The sky130 process will not reach 77 GHz. Not slowly, not expensively. Not at all. The IHP SG13G2 process is a 130 nm SiGe process with special high-speed transistors that reach about 350 GHz. This is the transistor type that real 77 GHz car radar uses. So the design splits into two chips, one on each process, with the boundary at the analogue-to-digital converter. One result of this belongs here and not in a footnote. You cannot prototype a 77 GHz front end on an FPGA. FPGA testing covers the digital half only. The high-frequency half is proven in silicon or not at all. AND ONE PLACE WHERE A FREE DESIGN KIT IS A PROBLEM, NOT A BENEFIT Everything above is true of the factory process. It is not yet true of the parts library. A paid 130 nm design kit comes with a tested library of analogue building blocks: reference circuits, protection cells rated to a stated level, memory. A free kit does not. So the ± 15 kV protection ring on chip 5, the temperature-stable reference on chip 3 and the memory on chip 1 are targets with no tested example to copy. We have to build and prove each one in silicon before it becomes a datasheet number. That is real work and real schedule. It is a bigger technical risk than the 200 MHz question, and §14 carries it as its own line rather than hiding it inside "design".

4.1 And what this process cannot do

The limits belong in the same section as the advantages. A reader who finds them later will assume we hid them. At 130 nm and 200 MHz we cannot build a high-performance AI accelerator, object recognition at video speed, a central car computer with multi-gigahertz processors and fast memory, or drone swarm control. Those all need much smaller processes. A 28 nm process gets you some of them. None of them can be claimed at 130 nm at any clock speed. That is exactly why the D100 drone chip is on a separate track using a 28 nm process and separate money. It is also why chip 9 says on its own face that it handles the local layer of a car's electronics and not the central computer.

The strategy draft states "10 new SKUs every year" and "over 1,000 active SKUs by Year 5". Those are inconsistent by 20x: $10/\text{yr} \times 5 \text{ yr} = 50$.

50 SKUs on mature nodes is already an aggressive catalogue for a company this size. 1,000 would need 200 tapeouts a year. Pick one number before this goes to an investor.

Shown below: the 50-SKU ramp, which is what "10 per year" actually produces.

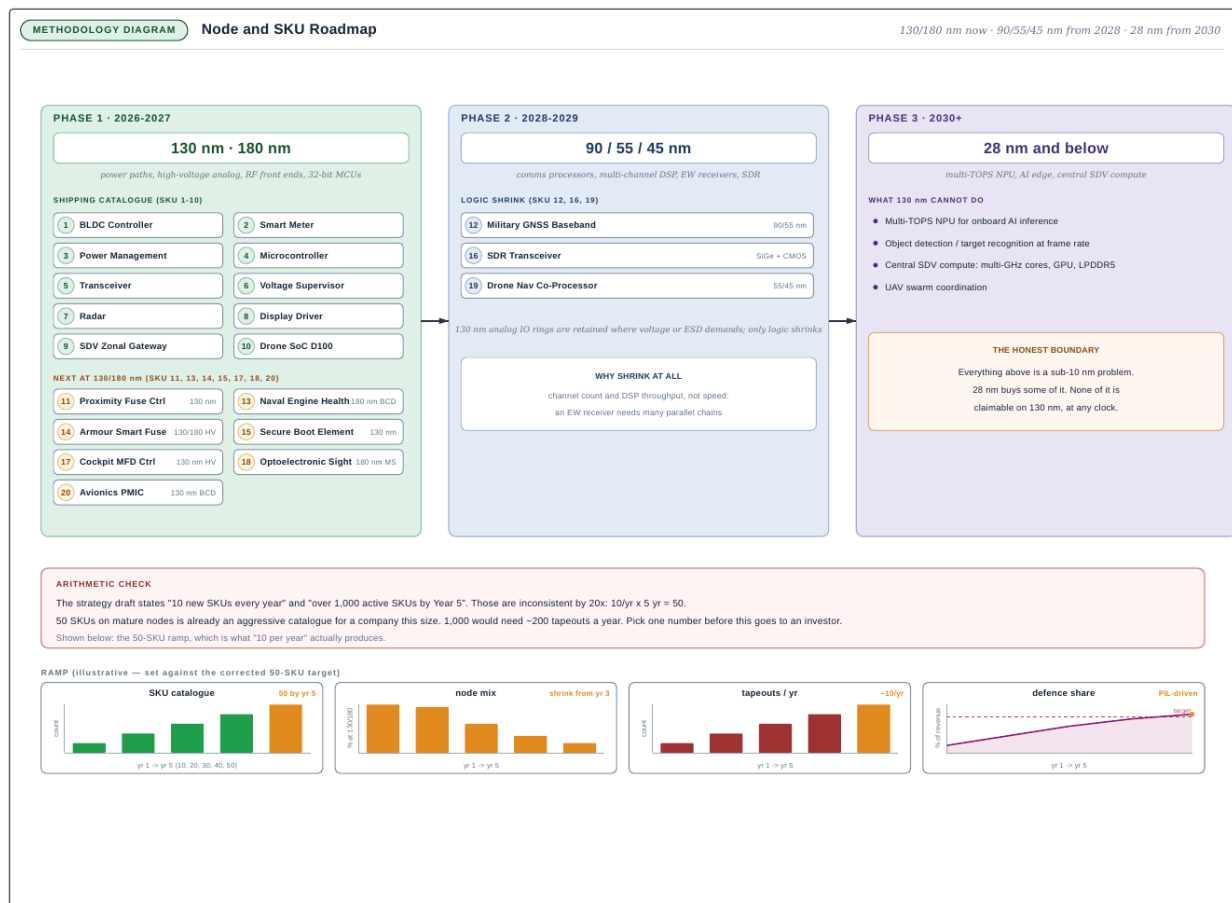


Figure 4. The process plan. 130 and 180 nm now. A move to 90, 55 and 45 nm from 2028, where the limit is the number of channels rather than the clock speed. 28 nm from 2030. The panel also corrects an arithmetic mistake in an earlier internal draft, because that draft's "1,000 chips" and "10 per year" could not both be true.

The process is right and the cost is right. What turns those two facts into a range of products, rather than one product, is the loop.

5 One loop every 198 days

Progress here is not measured in milestones. It is measured in factory runs. Our hiring plan, our cash need and how fast the product range grows all follow from how long one run takes, and from which half of it can be repeated.

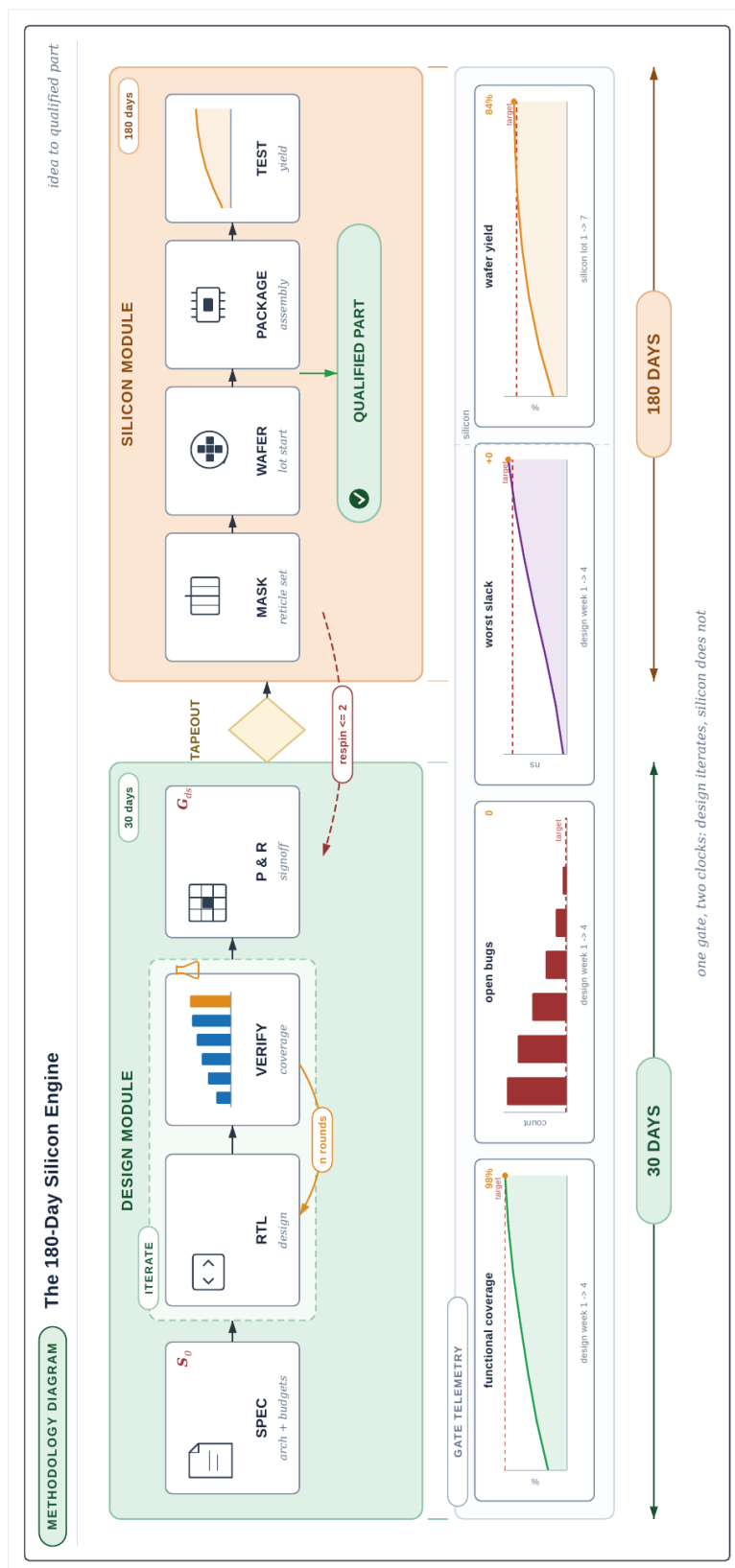


Figure 5. The 180-day engine. The design half cannot. The silicon half cannot. The small charts along the bottom are example shapes against stated targets, and they say so on their own face.

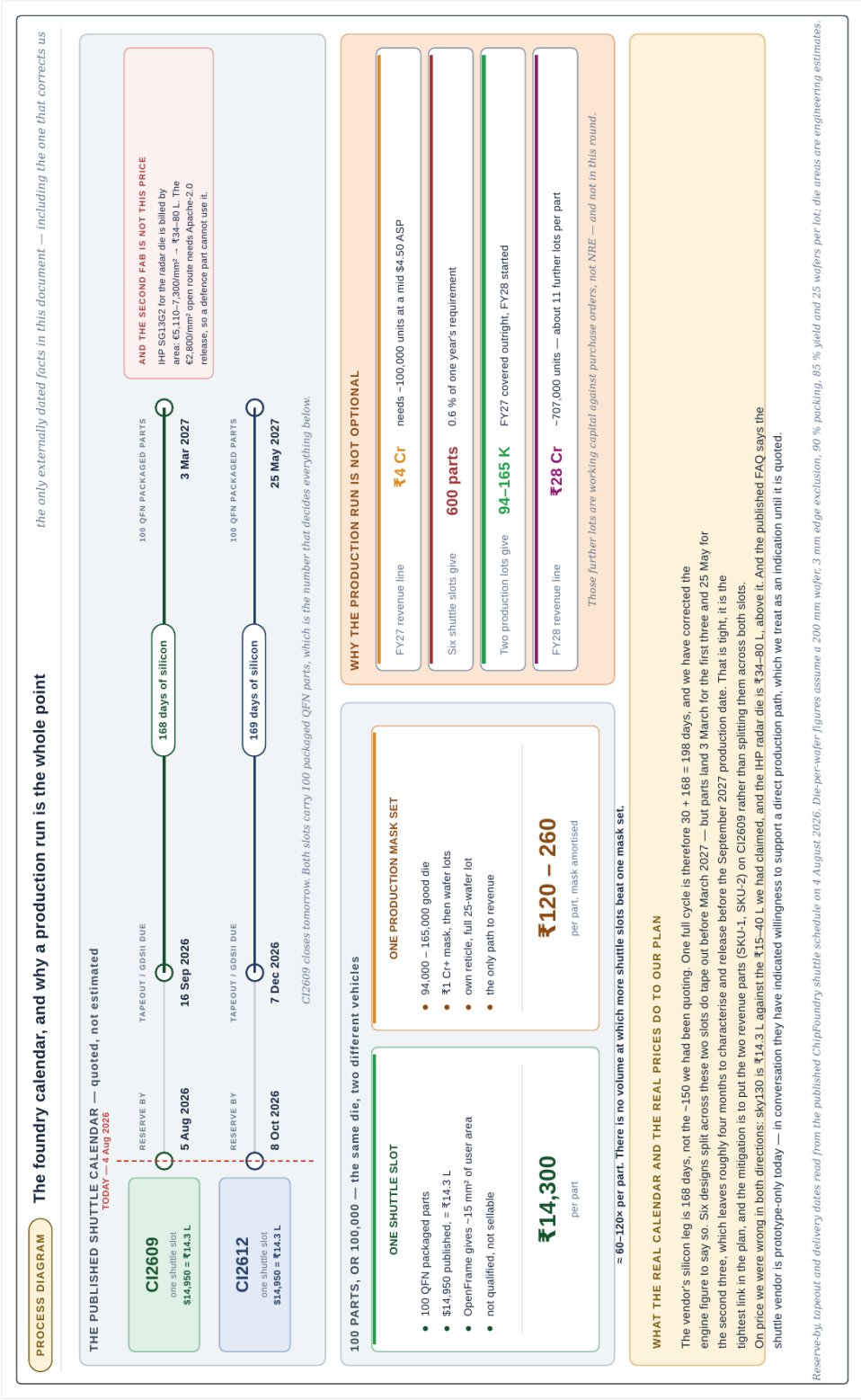
Design: 30 days or less. Specification, then circuit code, then verification, then layout, then final checks. Verification and circuit code go back and forth with each other. Nothing else in the chain does. This half is limited by how many engineers we have, and engineers are what money buys. The manufacturing gate. To go ahead you need enough verification coverage, no timing failures, and a clean layout check. These are targets today, not history, and the figures say so. Silicon: 168 days. Masks, wafers, packaging, testing. This half cannot be repeated at all. One mistake costs a whole cycle. That is why we enforce the design-complete gate before the factory window, not during it. That 168 is not our estimate. It is the gap between sending the design and receiving packaged chips on the published factory calendar. From 16 September 2026 to 3 March 2027 on the CI2609 run. From 7 December 2026 to 25 May 2027 on CI2612. We had been saying “about 150 days”, which was a rounded guess. So one full cycle is $30 + 168 = 198$ days, not 180. We would rather correct our own headline than have you find the factory's calendar and correct it for us. We keep the name because the design half is the part we control. The arithmetic is in Figure 6.

5.1 There are two speeds here, not one

The 30-day figure is for digital chips. It would be misleading to let it stand for the whole range. A digital block on an automated flow really does finish in weeks. An analogue block does not, because you cannot check a reference circuit or a protection ring on a computer. You need silicon in your hand. Mostly digital Mostly analogue Chips 1·2·4·8·9 3·5·6·7 Design loop

30 days or less to design files

one factory run per silicon-checked attempt Attempts needed limited by verification, on a computer usually 2–3, each needing silicon Realistic time to a tested chip one 198-day cycle plus 6–9 months two to three cycles plus approval §7 says the same thing from the other side when it says every analogue block has to be redesigned rather than moved. We would rather the two statements agree openly than have you find the gap and doubt both. It is also why chips 1 and 2, the two mostly-digital commercial ones, go first and carry the early revenue.



5.2 Who actually makes the production batch

The right-hand side of Figure 6 only works if somebody will make a dedicated mask set and a full batch of wafers. That needs stating carefully rather than assuming. The published position is prototypes only. Our factory service's own FAQ says the programme is “focused on prototype volumes (100 units per design)”, with production volumes planned “based on market demand”. Read literally, that means the production masks and first wafer batches that ₹2.26 Cr of this round is meant to buy had no named supplier at all. That would be a bigger hole than the packaging gap in §10. In conversation they have said they are willing to support a direct production route. That is a real change and it is why this is no longer just a hope. It is also not a contract, a price quote or a schedule, and we will not present it as one. The next thing to get is a written price quote for a production mask set and a first wafer batch. That is action 3 of seven in §14. One result of this is uncomfortable and belongs here. If the same supplier handles both the prototype and the production route, then our dependence on one supplier gets worse, not better. It now covers the whole path from design to volume. The backup is unglamorous but real. The sky130 process is SkyWater's own production process, so a direct factory relationship is possible. And SCL Mohali is the Indian backup.

5.3 The boundary we would rather state than be caught on

180 days ends with silicon in your hand, not with a product you can sell. Commercial chips ship 6–

9 months after that: characterisation, datasheet, test program development, approval batches, customer

design work. A screened defence chip adds another 9–12 months of MIL-883, JSS and CEMILAC checks, and that clock is not ours to speed up at any price. We say this because the alternative is a schedule that slips in front of you later. We budget ₹40–80 L per chip for approval as a separate line rather than hiding it in overheads, and §10 explains what that money actually buys. A 198-day loop gives us one chip per cycle. It does not explain how a small team builds fifty chips. That comes from what builds up between the loops.

6 How a product range builds up: blocks, not chips

We are one of about 500 designs on a single TinyTapeout die. That is a real credential and it points at a real strategy. But the strategy only survives a chip investor's questions if we change one word first, and we would rather change it ourselves.

6.1 What a shared die can and cannot do

TinyTapeout puts hundreds of small designs on one die behind a shared wrapper. Each design gets a fixed tile of space, and a switch picks which one connects to the shared pins. Only one design works at a time. That is excellent for what it does, and completely unable to do four things:

- Space. A tile is about 0.02 mm². That is a few hundred to a few thousand logic gates. A motor controller

with a power converter, a 16-bit converter, seven output channels and a processor is several mm². No sellable chip fits in a tile.

- Analogue and high voltage. The wrapper is digital only. No high-voltage devices, no reference circuits,

no converters, no thick output ring, no high-speed radar transistors. Chips 3, 5, 6, 7 and the output stage of chip 8 cannot be tested this way at all.

- Using many pins at once. The pins are shared and time-switched. You cannot test 3,840 display out-

puts, or seven motor channels, or four radar receivers. You can show a digital block works. You cannot measure noise, power supply rejection, or accuracy across temperature from a tile.

- Datasheet measurements.

6.2 The change of word

The mistake is calling a tile a chip. It is not. It is a building block. And every chip in our range breaks down into digital building blocks, and those do fit in a tile: the motor control datapath, the tuning loop, the meter filter, the fault table, the noise filters, the lockstep comparator, the display timing generator, the network shaper, the encryption cores, the memory controller, the radar processing control. Do not say “1,000 chips in five years.” A chip means a datasheet, an approval, a test program and a place on an approved-vendor list. A thousand of those cannot be built. A chip investor will reject it immediately, and that then makes every other number in the document look careless. Say “1,000 silicon-tested building blocks, combining into about 50 approved chips, 10–20 of them in volume.” Same ambition. Completely believable. And it explains how the chips get built quickly, which the other version does not.

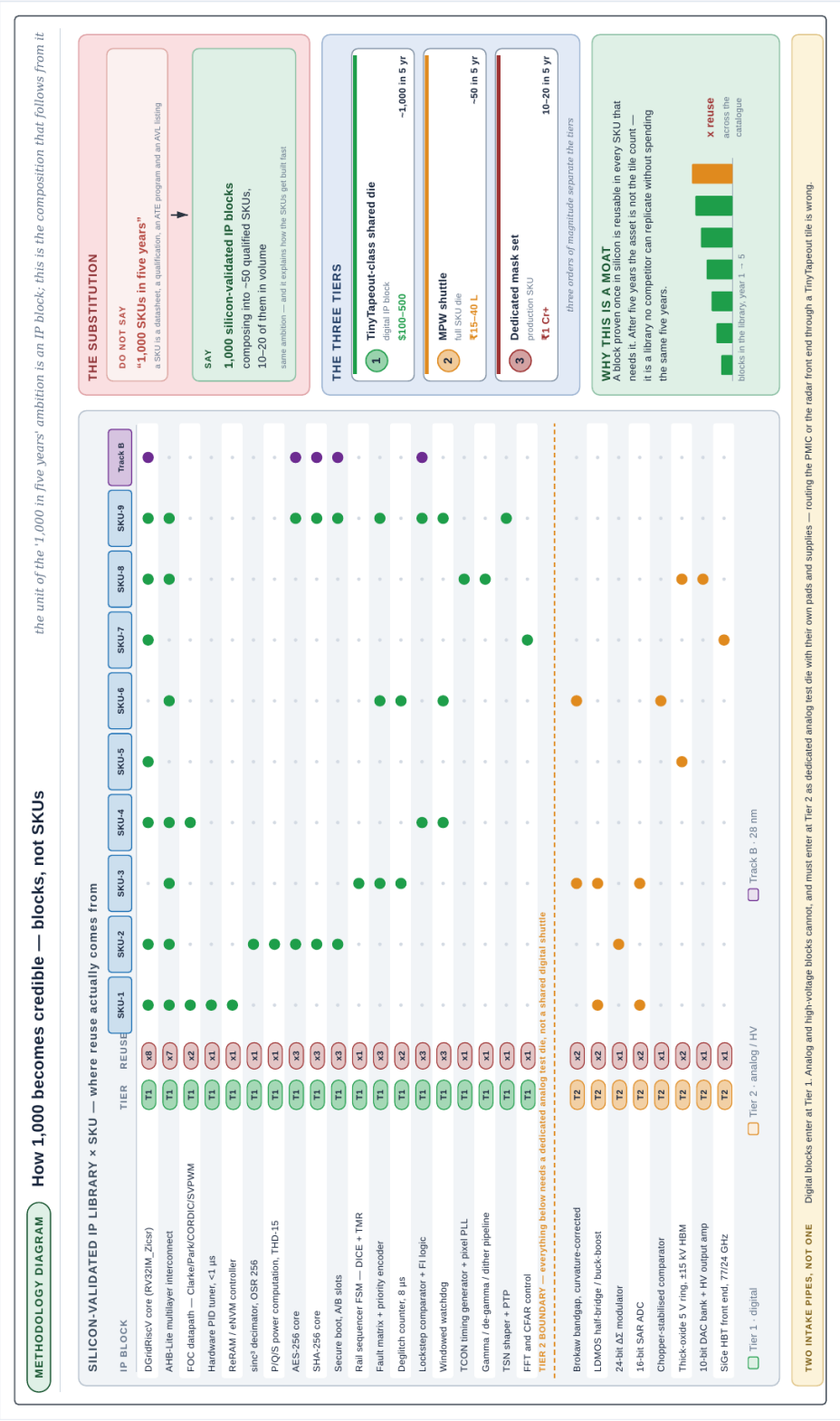


Figure 7. What follows from that. Twenty-seven building blocks against ten chips, with how many times each block is reused, and the analogue boundary drawn clearly. Everything below the dashed line needs its own analogue test chip and cannot go on a shared digital die at all.

Level Vehicle Unit Cost each What it proves 5-yr count shared die, TinyTapeout type digital block \$100–500 the block works in silicon 1,000 shared factory run full chip ₹14.3 L sky130 · ₹34–80 L IHP the chip meets its spec 50 dedicated mask set production chip ₹1 Cr+ we can ship it in volume 10–20 The three levels are a thousand times apart in cost, and that is the point. You spend a few hundred dollars to learn what would otherwise cost ₹1 Cr to learn. There are two entry routes, not one. Digital blocks enter at level 1. Analogue and high-voltage blocks cannot, and must enter at level 2 as their own test chips with their own pins and power supplies. Any plan that puts the power management chip or the radar front end on a shared tile is wrong. Saying so protects everything else here.

6.3 The precedent every informed reader will think of

A free design kit, a shared die and a range of small chips is not a new idea. The closest attempt ran at large scale. The shared-run programme that made sky130 available produced several hundred manufactured designs over four years with major corporate funding. It did not produce a commercial range of approved chips. Anyone who knows this field will think of it within thirty seconds of reading this section. So we would rather say what we think is different than pretend it did not happen. Three differences. Weigh each one yourself:

- Those runs were nearly all level 1 in our own scheme. Blocks and demonstrations on a shared wrap-

per, which is what a shared wrapper is good for. Almost none went on to a full chip run, and essentially none went to a dedicated mask set. That programme proved blocks and then stopped, because proving blocks was its purpose. Levels 2 and 3 are the part nobody has demonstrated on a free design kit, including us, and §14 says so.

- They had no protected demand. A hobby or research chip has no approved-vendor list waiting for it

and no import ban creating a buyer. The legal position in §1 is what turns the same technical approach into a business, and it did not exist for them.

- We are not selling the factory run. We are selling the chip. A run service earns money from the

number of designs. We earn money from approved chips, which is a much harder target and a much smaller number. Fifty, not a thousand. If those three do not convince you, then the honest answer is that this is the weakest part of the plan on past evidence. That is exactly why §14 says the round has to prove levels 2 and 3, not level 1.

6.4 Why this is a real advantage and not just a cost trick

A block proven once in silicon can be reused in every chip that needs it. After five years the asset is not the number of tiles. It is a library of silicon-tested blocks that a competitor cannot copy without spending the same five years. The lockstep comparator tested for chip 4 is the same block that makes chip 9's safety section and the drone chip's backup system believable. That is how this compounds, and it is why the range can grow by several chips a year without several times the staff.

6.5 Two things we would argue with in our own plan

Custom chip-to-chip connections stop us selling a single die. We chose not to use the UCIe industry standard. That avoids a real area and power cost, but it means no other company can buy one of our dies as a component. If selling dies is ever the plan, the connection standard has to be chosen now. Changing it later is a redesign. Packaging is a supply chain that is not yet in our plan. The end goal is six chips in one package: power,

motor drive, communications, safety, control and analogue, with the 28 nm AI chip joining later. This multiplies volume, because one packaged product contains six chips. But it needs an assembly partner, package-level testing, and an answer to what happens when one chip in six is faulty. That last question forces you to test every chip before assembly, and \$10 carries it as a cost and a schedule line rather than a detail. One method, one library. The next question is which factories it runs in, and whether "works on any process" survives contact with an analogue circuit.

7 One method, three factories

One design method, three factories, built to move between processes. The three are not alternatives. They are a sequence with three different jobs. Stage Factory / kit What it gives Status
Proven sky130 · SkyWater, USA the finished 130 nm test chip on our own flow Done Different
SG13G2 · IHP, Germany free 130 nm SiGe process with 350 GHz transistors — 77 and 24 GHz
radar, fast data links Planned Indian 180 nm · SCL Mohali, India military-approved factory:
design, manufacture, screening and approval all in India Planned sky130 is where the method is
proven and where digital chips get cheap first silicon. IHP is the only free design kit that can make
the radar chip at all, and nothing else in the range needs it. SCL is the answer to the question a
defence customer always asks. That question is not "does it work" but "where is it made". And it
is planned, not done. We keep the three-stage description because it does not pretend otherwise.

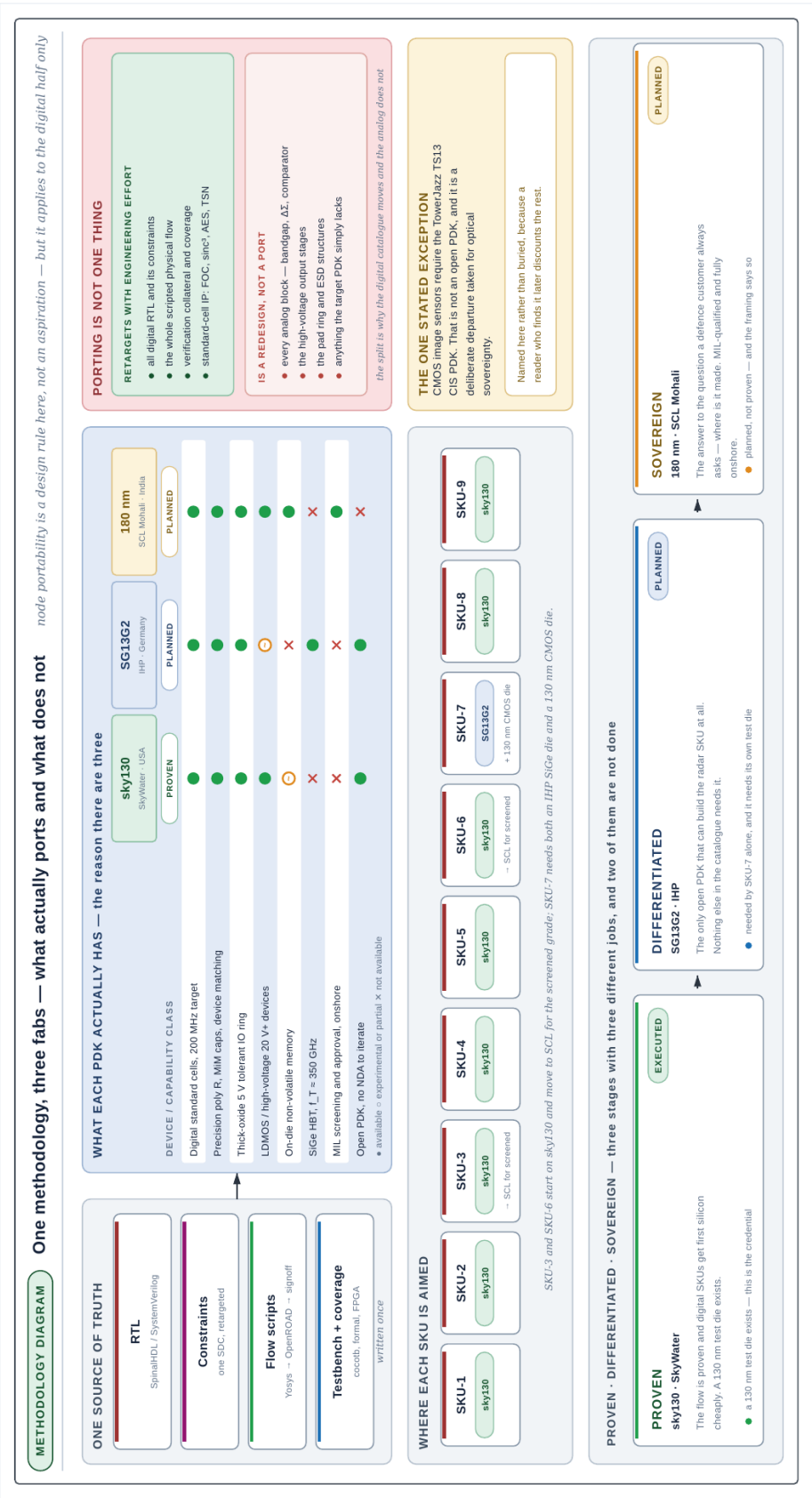


Figure 8. What each design kit actually contains, and what moving a design really costs. The digital half moves with engineering effort, because the flow is scripted and the kit can be swapped. Every analogue block, the high-voltage stages and the pin ring are a redesign, not a move. That split is the honest meaning of “works on any process.”

One stated exception to using free kits. Camera sensor chips need the TowerJazz TS13 kit, which is not free. It is a deliberate, limited exception taken so India can make its own camera sensors, and we name it here rather than burying it. Three factories, one flow, one library. Seven of the ten chips also share one processor, and standardising on it is what lets a small team build a whole range.

8 The processor every chip shares

Seven of the ten chips contain a processor, and it is the same processor each time. That is a deliberate restriction. The software tools, the start-up code, the debug method, the interrupt system and the verification work all get written once and spread across the whole range. DGridRiscV is our product version of the VexiiRiscv processor generator. What follows is read out of the circuit code our tools actually produced, not from a wish list. Every row can be checked against that file. Property Value How we know Instruction set RV32IM_Zicsr (misa = 0x40001100) the constant in the generated code Privilege level machine mode only PrivilegedPlugin Issue one instruction at a time, in order one lane0 in the code Pipeline

2 fetch stages, 6 execute stages

fetch_logic_ctrls_0..1, execute_ctrl0..5 Multiply and divide in hardware MulPlugin + DivPlugin Shifter full barrel shifter, one cycle BarrelShifterPlugin Memory protection PMP regions PmpPlugin Instruction fetch no cache, 32-bit AXI4 port FetchCachelessAxi4Plugin Data access no cache, 32-bit AXI4 port LsuCachelessAxi4Plugin Branch prediction none no BtbPlugin or RasPlugin Floating point none no FPU; the decode signals are tied off Compressed instructions none — 4-byte instructions only not set in misa

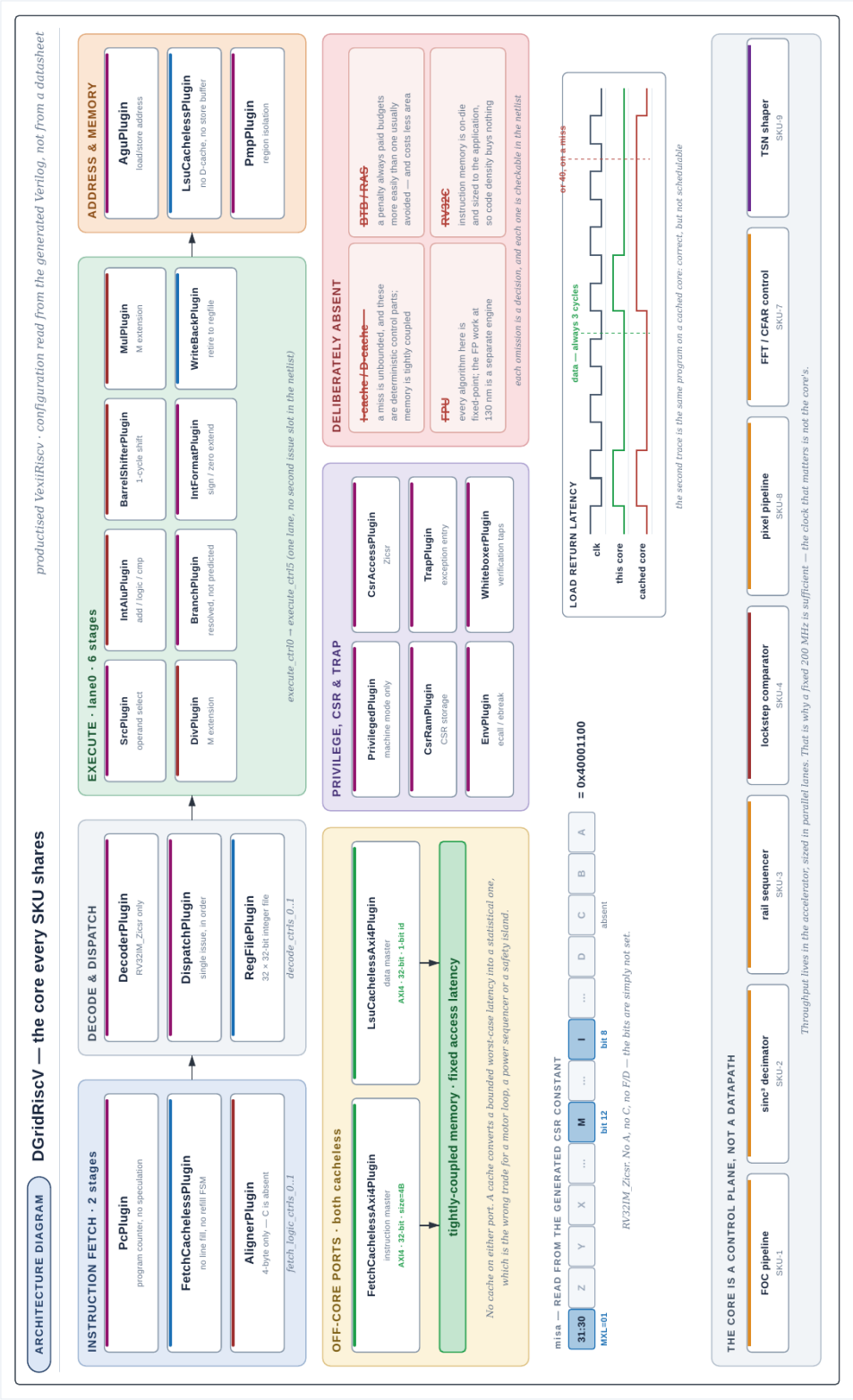


Figure 9. The processor as it is actually built. Read the pink panel first. Four features are deliberately missing, each with its reason, and each can be checked in the generated code.

8.1 Every missing feature is a decision

A reviewer will read the second half of that table as a list of gaps unless we explain each one. No caches. A cache is a small fast memory that keeps recently used data. It usually helps, but sometimes it misses and then the access takes far longer. These are control chips that must respond in a guaranteed time. A cache turns a guaranteed worst case into an average, and that is the wrong trade for a motor loop, a power sequencer or a safety monitor. Our processor goes straight to closely attached memory with a fixed access time. Chip 1's hardware control loop responds in under 1 microsecond, and that number only means something because nothing in the path can miss. No branch prediction. Same reason, plus chip area. A cost you always pay is easier to plan for than one you usually avoid. And these control loops are short and dominated by data processing rather than branching. No floating-point unit. Every calculation in our range uses fixed-point maths by design. Fixed-point is smaller, faster, and gives exactly the same answer on every chip. That matters when the same calculation must match between a screened chip and a commercial one. Our floating-point work at 130 nm sits on the test chip as a separate engine, not inside the control processor. No compressed instructions. Compressed instructions save memory space. That matters when instruction memory is off the chip and expensive. Here it is on the chip and sized to the job, and removing the feature removes an alignment case from the fetch path. Memory protection instead of a full memory manager. These chips run bare-metal code or a small realtime operating system. What they need is to keep the secure start-up code, the application and the safety monitor separate from each other. Memory protection regions do exactly that, at a fraction of the chip area a full memory manager needs.

8.2 Why 200 MHz is enough, and what the number is not

Because the processor is a controller, not a calculator. In every chip the heavy work is done by dedicated hardware: the motor control pipeline, the meter filter chain, the display pixel pipeline, the radar processing. The processor sets that hardware up, watches it, handles faults and talks to the host system. The clock speed that matters is the dedicated hardware's, and that hardware is sized by how many parallel lanes it has, not by megahertz. Which means 200 MHz is a target for the processor, not a promise for every chip, and we have been using the number too loosely. Three points go with it. It applies to the DGridRiscV processor and the bus around it. Several chips do not need it at all: chip 6 is a comparator with a watchdog, and chip 5 is a line driver whose limit is signal timing, not clock speed. And the chips that do want it will each reach their own speed, because a 3,840-output display driver and a six-channel meter front end are not the same timing problem. What we will publish is a measured speed for each chip once the analysis is run, not one number for all nine. That is the machine: the process, the loop, the library, the factories and the processor. What follows is what we build with it.

PART FOUR

Ten chips, and what it takes to turn a die into a product Each chip exists because a specific socket is filled by an import today. Getting from working silicon to something a customer can buy is a separate problem, and it is where hardware plans usually break.

9 Ten chips, and what each one is for

Chip Product Process Why it exists BLDC motor controller 130 nm two-chip driver-plus-processor sets combined into one Smart-meter chip 130 nm 250 M meter rollout; tamper detection is required in tenders High-reliability power chip sky130 → SCL 180 nm

28 V power rails in every avionics box

Lockstep microcontroller 130 nm the fault-tested processor, made into a product Interface transceiver 130 nm replaces discontinued parts; highest attach volume Voltage supervisor sky130 → SCL simplest chip — goes through approval first 4D radar front end IHP SiGe + CMOS the one chip with a hard physical process limit Display driver 130 nm high voltage named on PIL-5, BEL, due Dec 2027 Car zonal controller 130 nm local layer only — the central computer is out of scope Track B D100 drone chip TSMC 28 nm the high-performance one; separate round, revenue-gated Two ordering decisions are deliberate, not accidental. Commercial chips go first. The motor controller and the meter chip sell fastest and build credibility. The defence versions follow behind them on the same approved silicon, at prices 10 to 100 times higher. And the simplest chip goes through approval first. Chip 6 is a supervisor with one comparator chain and a short test list, so it carries the MIL-883 screening process for everything behind it. If that process holds a surprise, we would rather find it on a supervisor than on a four-rail power chip. The ten architecture drawings that follow each show the internal blocks, the timing behaviour, example performance curves, a market panel with volumes and prices, and where FPGA testing stops and the real chip begins. Volumes and prices are our own estimates unless a source is named. Performance curves are example shapes against stated targets, not measurements.

9.1 Chip 1 · BLDC motor controller

What it is. A DGridRiscV processor at 200 MHz. A hardware control loop that responds in under 1 microsecond. A full motor-control calculation unit. Seven output channels for driving the motor. A 16-bit converter to read sensors. A 5–120 V power converter. And 256 KB to 1 MB of memory that keeps its contents without power.

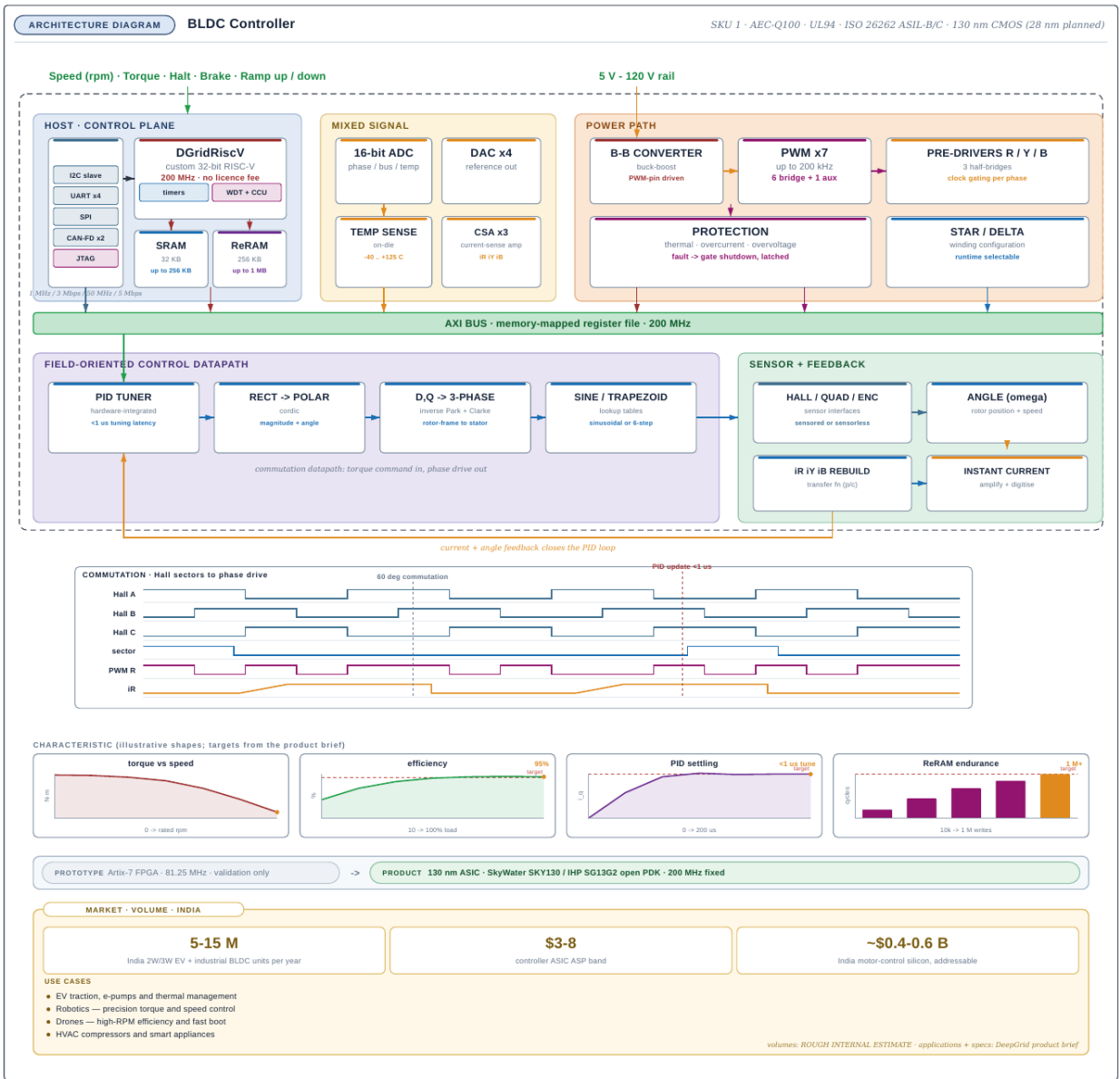


Figure 10. Motor controller, 130 nm. The processor and control section are on the left. The motor-control calculations are in the middle. The power converter and the output drive stage are on the same chip, which is the whole reason this product exists.

What it replaces. Today a customer buys two chips: a driver chip such as a DRV83xx, plus a separate microcontroller, plus the components between them. We put all of it on one chip. The motor wiring type can also be switched in software while running. Why this process. 5 to 120 volts works natively here. So the power converter, the output drive and the processor all sit together. A newer process would push the power circuits back onto the board and defeat the purpose. Where it sells, and how many. Guaranteed demand in every ASWA robot joint and every drone module we build. Commercially we estimate 5 to 15 million units a year across Indian two- and three-wheeler electric vehicles and industrial motors, at \$3 to \$8 each. Separately, the government energy-rating push into BLDC ceiling fans is worth another ₹250–550 Cr a year of driver demand. The legal or demand hook. PIL-5 lists BLDC motor with encoder for anti-tank missiles, due December 2025. That is a direct match, though the listing is for the whole motor unit and not the chip inside it. §11 explains why that distinction matters. Status. The FPGA version at 81.25 MHz is for testing only. The product is the 130 nm chip at 200 MHz. The non-volatile memory is planned using sky130's experimental option, with SCL's version as a backup. First factory run: cycle 1.

Anchor customer, named: Airgap Technology — an Indian maker of electric-vehicle motors and powertrains for two-, three- and four-wheelers — is the first named external buyer for this chip. **PLANNED**

9.2 Chip 2 · Smart-meter chip

What it is. A six-channel measurement front end accurate to 24 bits, with a filter chain that averages 256 samples. It calculates real, reactive and apparent power up to the 15th harmonic. It includes AES-256 and SHA-256 security engines. And it has an always-on clock section that uses under 2 microwatts.

Anchor customer, named: Ripple Metering, an Indian smart-meter maker working on LoRaWAN networks. The discipline in §11.2 still holds: this chip's production masks wait for Ripple's signed letter. **PLANNED**

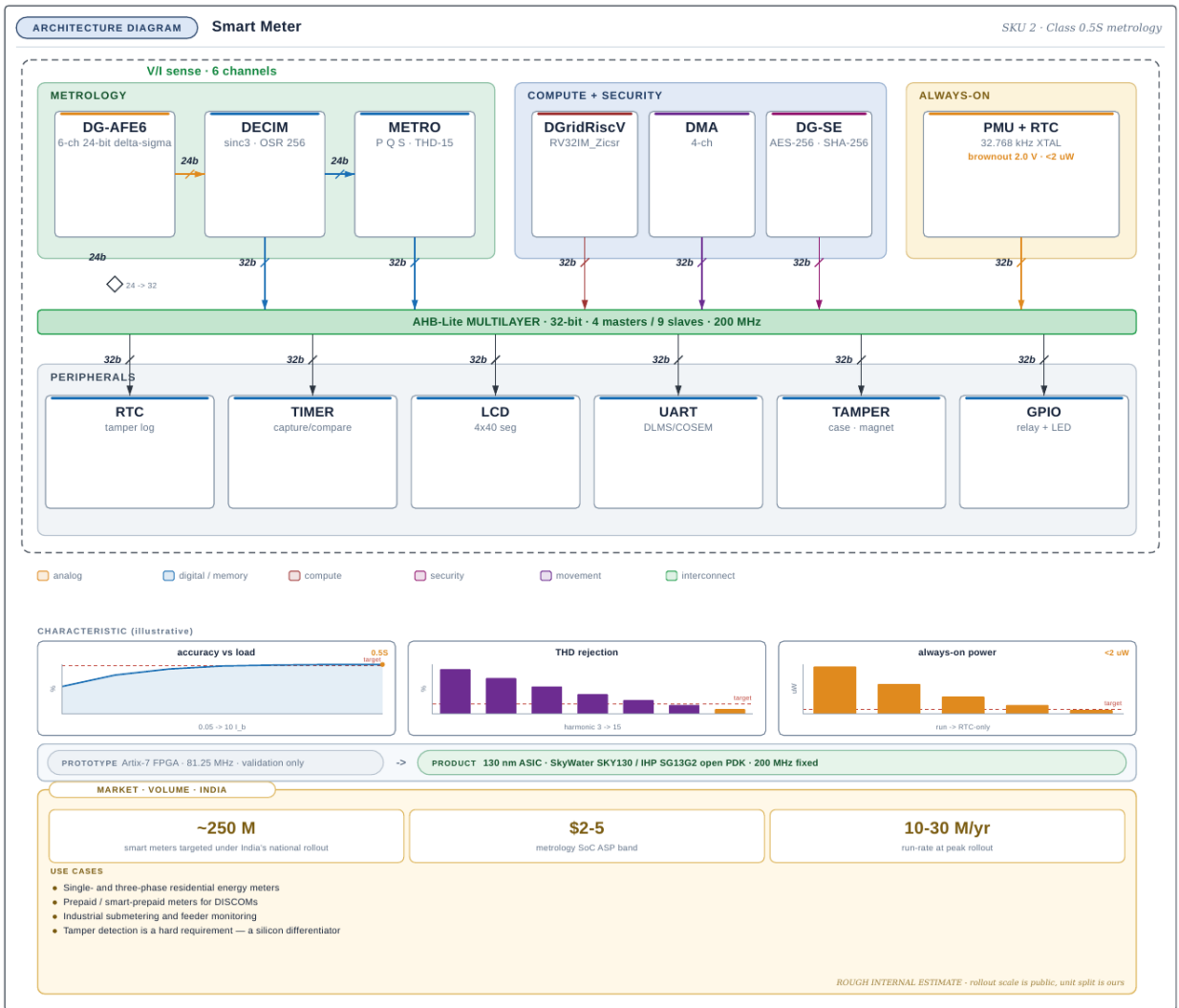


Figure 11. Smart meter, 130 nm. Class 0.5S accuracy. Six 24-bit measurement channels feed a filter, then the power calculation, then the secure and always-on sections.

What it replaces. A measurement chip such as an ADE9153 or V9203, plus a separate meter microcontroller. We combine them behind one internal communication network. Why this process. A 24-bit measurement circuit needs accurate resistors, good capacitors and closely matched transistors. 130 nm has all three. Where it sells, and how many. India's national programme to install about 250 million smart meters. We estimate 10 to 30 million units a year at peak, at \$2 to \$5 each. The legal or demand hook. Tamper detection is a requirement in meter tenders, not an optional feature. That makes it something the chip must do, not something software can add later. Status. The blocks are tested on FPGA. The product is a 130 nm chip at 200 MHz. Because a free design kit has no built-in flash memory, we use read-only memory plus error-corrected RAM plus encrypted external memory. A version using SCL's memory is planned for sealed meters. First factory run: cycle 1, alongside chip 1.

9.3 Chip 3 · High-reliability power chip

What it is. An input stage that meets the DO-160 and MIL-STD-461 aircraft standards. A first-stage converter with a stable control loop. Four power outputs that switch on in a controlled order and limit their own current. A temperature-stable voltage reference accurate to 12 parts per million per degree. A converter for reading telemetry. And a control state machine protected against cosmic-ray upsets using DICE latches and triple redundancy.

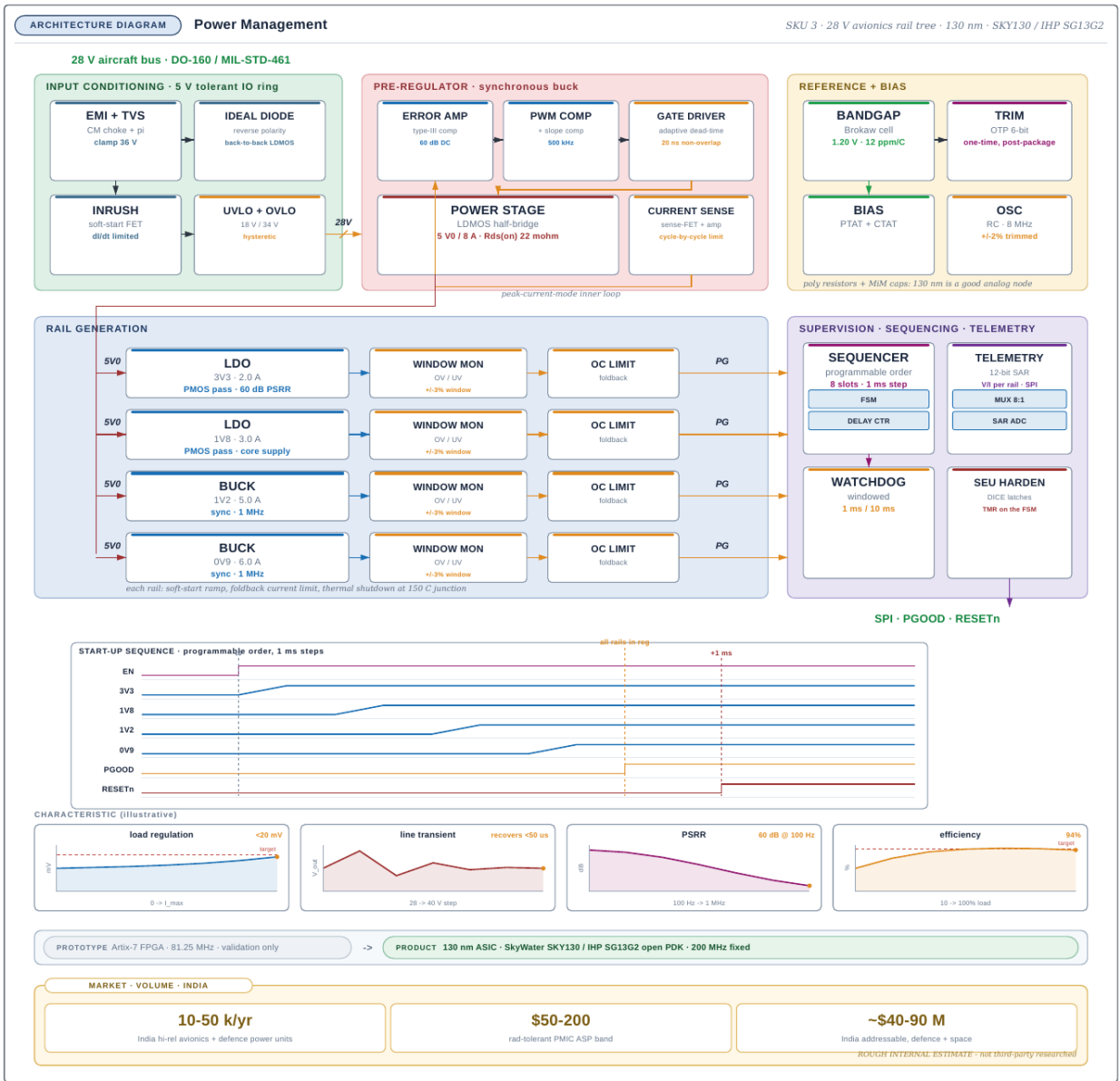


Figure 12. Power management, using sky130's 20 V devices, with a path to SCL's 180 nm process. An aircraft-standard input stage, a first-stage converter, four sequenced outputs with current limiting, and a radiation-protected control state machine.

What it replaces. Texas Instruments and Analog Devices military-grade power chips inside aircraft and armoured-vehicle boxes. This is genuine analogue design — a voltage reference, a compensated control loop, a telemetry converter — not a digital chip with a power stage bolted on. Why this process. High-voltage devices let a 28 V input stage and a 0.9 V processor supply sit on one chip. And radiation resistance comes from the transistor size itself, not from something added on. Where it sells, and how many. We estimate 10,000 to 50,000 units a year in Indian high-reliability power, at \$50 to \$200 each once screened. Small volume, 60–75 % gross margin, and once approved it stays on the vendor list for decades. The legal or demand hook. SRIJAN category NSG-5962. PIL-5 lists two tank power converters from BEL, 16–40 V input at 4 A, due December 2026 and December 2028. Status. MIL-883 screening process. Radiation protection by design only. We make no radiation-hardened claim beyond that.

9.4 Chip 4 · Lockstep microcontroller

What it is. Two DGridRiscV processors running the same program two clock cycles apart. If they ever disagree, comparators on the bus and at the output raise a fault signal within two cycles. Error correction is carried on every internal connection.

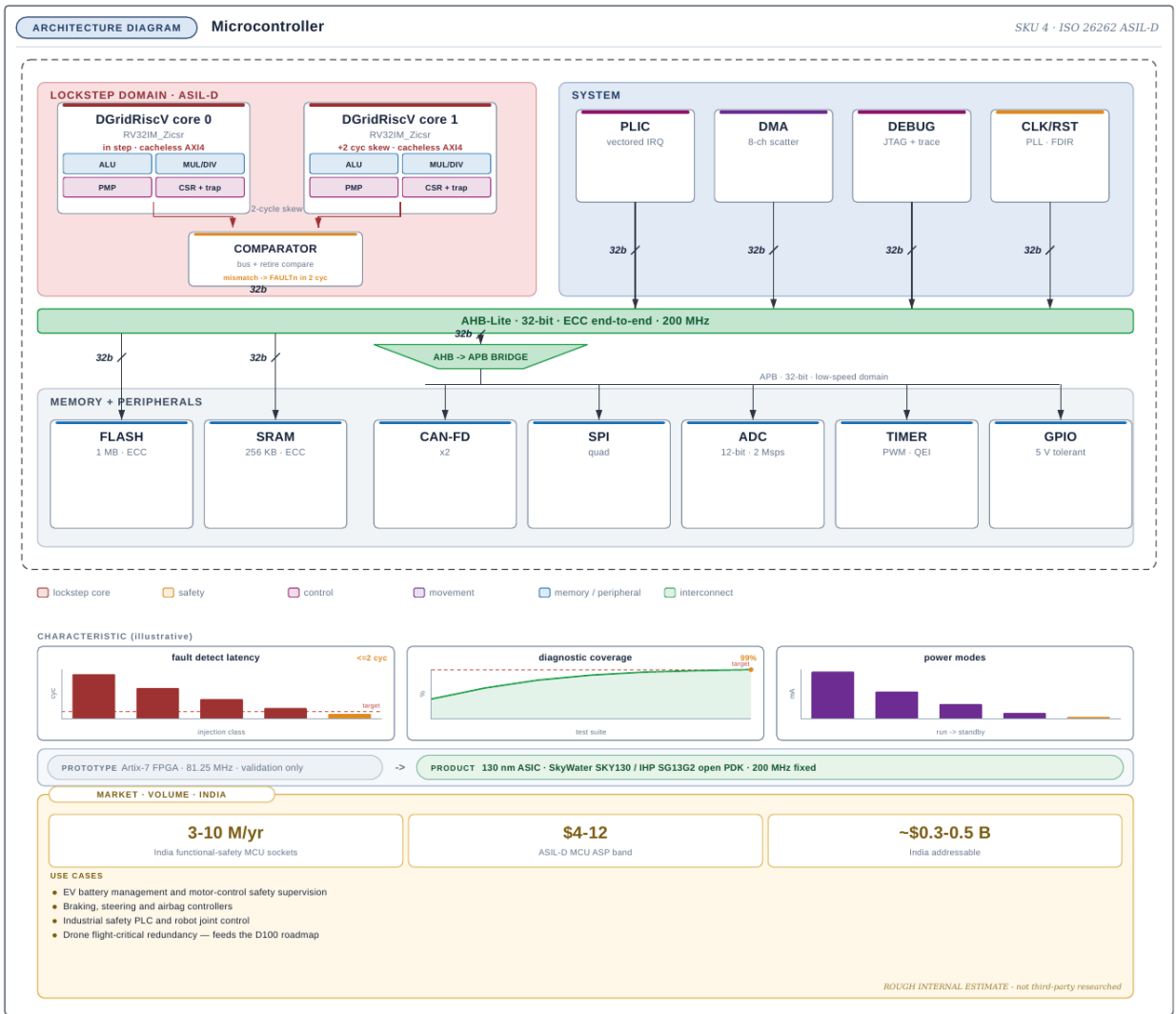


Figure 13. Microcontroller, 130 nm, on a path to ISO 26262 ASIL-D safety approval. Two processors running two cycles apart, compared at both the bus and the output.

What it replaces. Safety microcontrollers from Microchip and Renesas. We estimate 3 to 10 million units a year in India at \$4 to \$12 each. Why this process. Running two processors costs chip area, not speed. An older process makes that duplication affordable, and makes the radiation protection cheap enough to leave in. Where it sells, and how many. Electric-vehicle battery management, motor safety monitoring, braking and steering controllers, robot joints. It is also the control brain that sits next to chip 1, and the backup system feeding the drone chip. The standards hook. ISO 26262 ASIL-D is the highest automotive safety level. The circuit code already exists and has been fault-tested, carried over from the drone chip programme. Status. FPGA testing at 81.25 MHz, then a 130 nm chip at 200 MHz. The 1 MB error-corrected flash shown in the drawing needs SCL's memory. The first version on the free kit uses read-only memory plus RAM plus encrypted external memory.

Anchor customer, named: MCEME — and this one is not a prospect: ₹1.01 crore is already under contract (₹23.01 lakh delivered, ₹78.39 lakh L1). **DONE**

9.5 Chip 5 · Interface transceiver

What it is. Thick-oxide 5 V output stages. Receivers with 30 millivolts of noise margin. A fail-safe bias so the line reads correctly when nothing is driving it. And static-discharge protection to ± 15 kV.

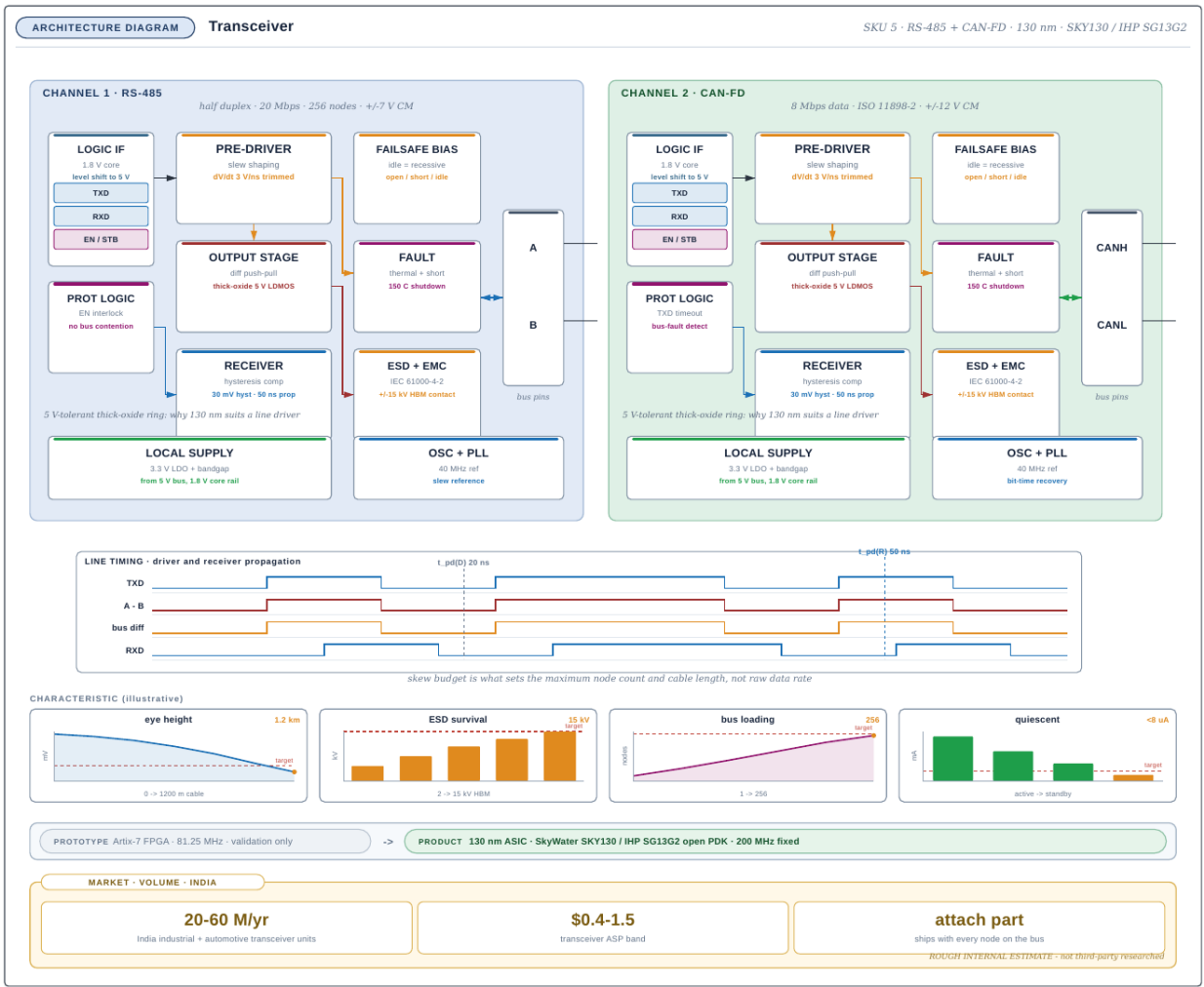


Figure 14. Transceiver, 130 nm, with a thick output ring. RS-485 and CAN-FD on one chip, with fail-safe biasing and a ±15 kV protection ring.

What it replaces. Interface chips from Texas Instruments, Analog Devices and Renesas. Two things drive this: the original part gets discontinued, and defence buyers face sourcing rules that block the import. Why this process. A newer process is actively worse for this kind of chip. A line driver needs a 5 V-tolerant thick output ring, and newer processes do not have one. Where it sells, and how many. We estimate 20 to 60 million units a year, because one transceiver ships with every node on every communication bus. \$0.4 to \$1.5 commercially, and 10 to 100 times that for screened versions. The legal or demand hook. Replacing a discontinued part is the fastest conversation available with a defence company, because the part is already approved into a system that can no longer buy it. What they need is an exact electrical match, not a new design win. Status. Signal timing, not raw data rate, decides how many nodes and how long a cable can be, and the design is budgeted for that. Common-mode behaviour follows ISO 11898-2. A high-voltage fault-tolerant version runs on SCL or IHP. Second factory run.

9.6 Chip 6 · Voltage supervisor

What it is. Comparators that cancel their own drift. Resistor dividers matched to 0.1 %. An 8 microsecond filter to ignore switching noise. A temperature-stable reference accurate to 10 parts per million per degree. And a watchdog timer that checks the processor is still alive.

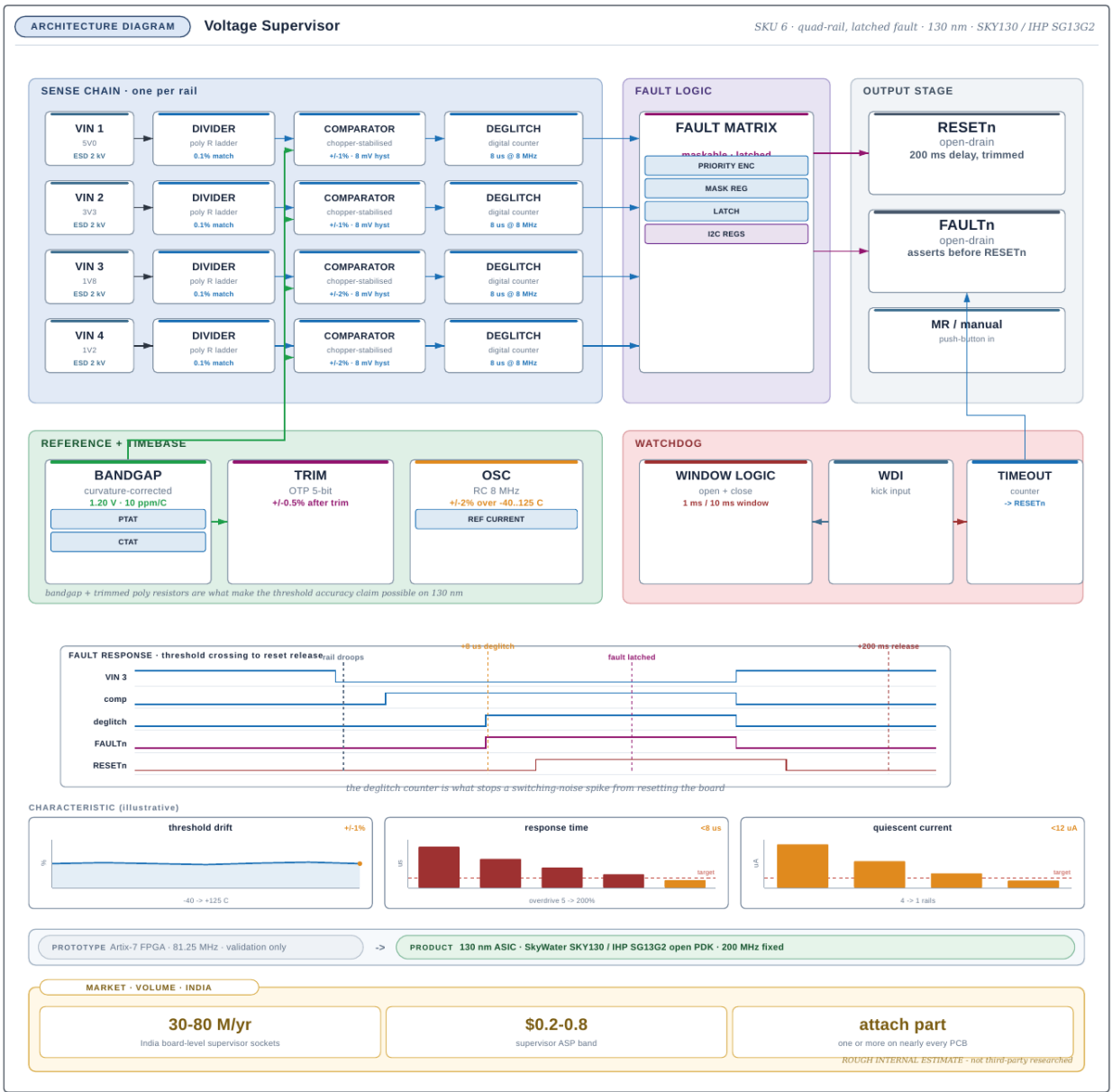


Figure 15. Voltage supervisor, sky130 with a path to SCL 180 nm. Drift-cancelling comparators against a temperature-stable reference, an 8 microsecond noise filter, and a watchdog timer.

What it replaces. Supervisor chips from Texas Instruments and Maxim. We estimate 30 to 80 million units a year, because nearly every circuit board has one or more. \$0.2 to \$0.8 commercially, and 10 to 100 times that screened. Why this process. Accuracy from -40°C to $+125^{\circ}\text{C}$ comes from a trimmed reference and matched resistors. Both are strengths of an older process. Where it sells, and how many. Every avionics box and every drone module. This chip has the highest number of sockets in our whole range. Why it goes first. It is the simplest chip we build, so it carries the MIL-883 screening process for everything behind it. §10 explains that decision. Status. The noise filter is the design story: ignore switching noise without slowing a real fault response below 8 microseconds. First or second factory run.

9.7 Chip 7 · 4D radar front end

What it is. 77 and 24 GHz radar, using frequency-sweep detection. Two transmit and four receive channels. A high-frequency SiGe chip for the radio part, and a 130 nm CMOS chip for the digital part. The boundary is at the analogue-to-digital converter.

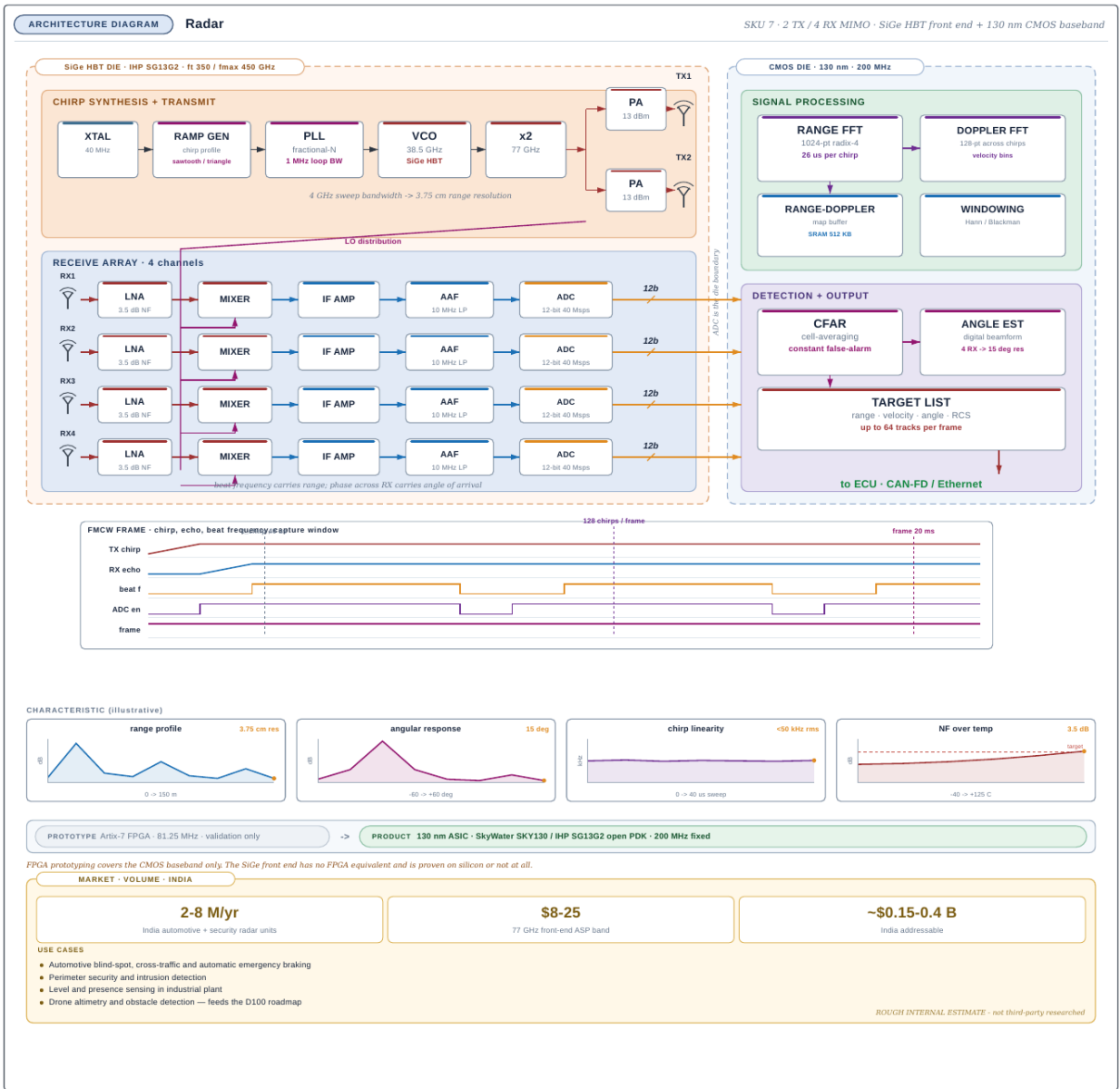


Figure 16. 4D radar, split across an IHP SG13G2 SiGe chip and a 130 nm CMOS chip, with the boundary at the converter. The SiGe side generates the radar sweep, distributes the local oscillator and holds the receive chains. The CMOS side does the range and speed calculations, target detection and beam forming.

What it replaces. Imported 77 GHz radio front-end chips and the digital chips that go with them. Why this process, and the hard limit. This chip cannot be built on ordinary CMOS. The IHP process provides the high-speed transistors that real 77 GHz radar uses. This is the one place in our range where the process choice is physics, not economics. Where it sells, and how many. We estimate 2 to 8 million units a year in Indian car and security radar at \$8 to \$25 each, plus our own AD2 platform and defence radar from FY28. The legal or demand hook. The defence ministry lists radar warning receivers for Su-30 MKI and Mi-17 aircraft, weapon-locating radar, battlefield surveillance radar at 10 and 30 km, shipborne 3D radar and precision approach radar. Status, and its own risk. FPGA testing covers the digital half only. The high-frequency half needs its own test chip. This chip must not be scheduled as if FPGA testing has reduced its risk.

9.8 Chip 8 · Display driver

What it is. LVDS and MIPI DSI video inputs. A colour processing chain: reverse gamma, colour correction, gamma, then dithering. A timing controller with its own pixel clock generator. And a bank of 10-bit converters feeding high-voltage output amplifiers across 3,840 column outputs.

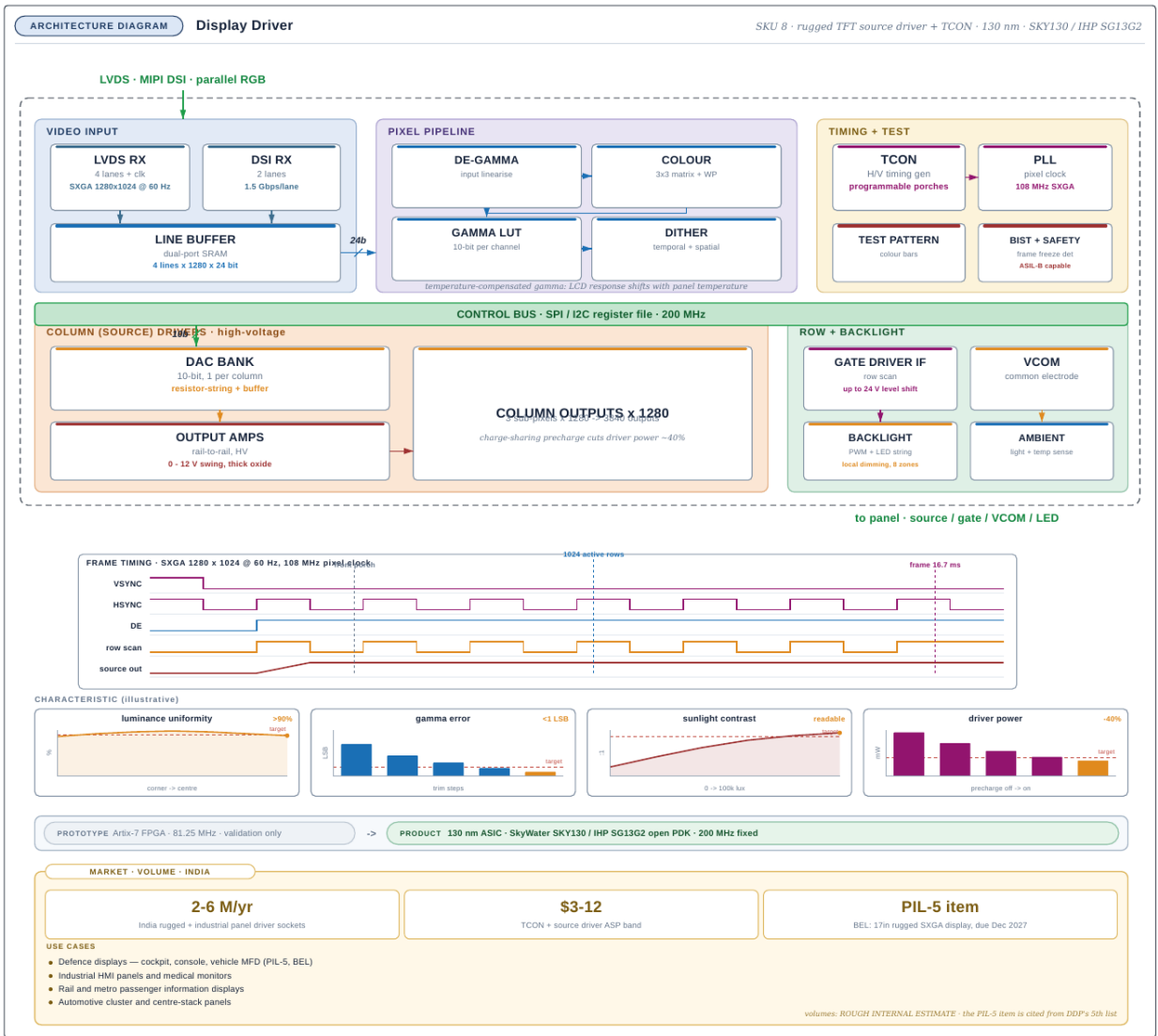


Figure 17. Display driver, 130 nm high-voltage CMOS. Video input into the colour chain, a timing controller with a pixel clock, and a bank of 10-bit converters driving 3,840 high-voltage columns.

What it replaces. Imported timing-controller and column-driver chip sets in rugged and industrial display panels. Why this process. Column drivers need to swing 0 to 12 volts into a display panel. That is a thick-oxide, high-voltage problem, which is exactly what an older process is for. Where it sells, and how many. We estimate 2 to 6 million units a year in Indian rugged and industrial display sockets at \$3 to \$12 each. The legal or demand hook. PIL-5 lists a 17-inch rugged 1280 × 1024 display from BEL Defence Electronics, due December 2027. This is the closest thing in the lists to a direct hit on one of our chips. Status. The cockpit display version is the same chip with a wider temperature colour table and a highervoltage output ring. That is the reuse argument made real.

9.9 Chip 9 · Car zonal controller

What it is. A lockstep safety section. A full hardware security module to the EVITA standard. Secure start-up with two software slots so an update can be rolled back. An over-the-air update engine. A timesensitive network switch with precise time synchronisation. Support for six different in-vehicle network types. And sixteen electronic fuses.

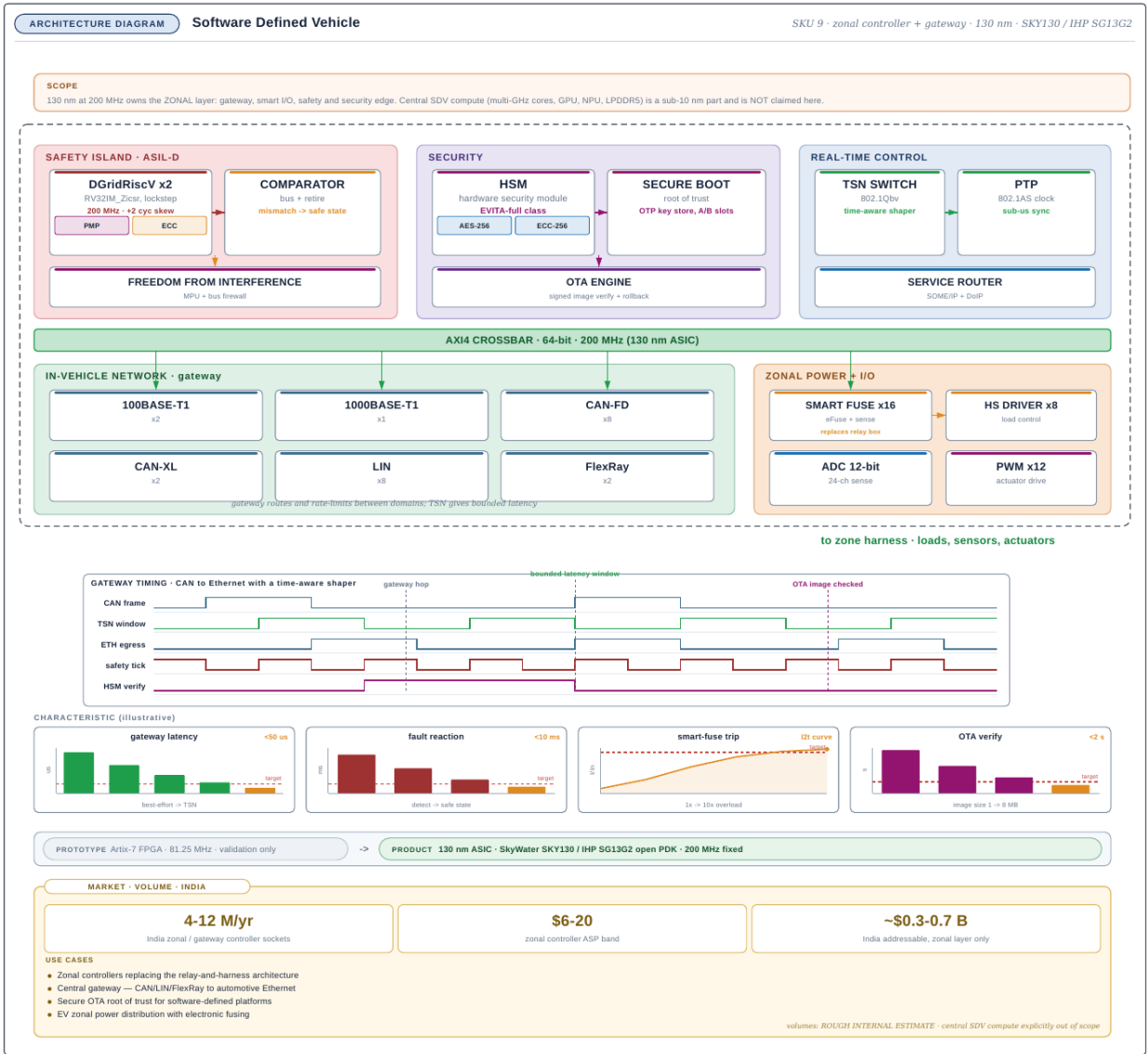


Figure 18. Car zonal controller, 130 nm. An ASIL-D safety section, a full hardware security module, a time-sensitive network switch, six network types and sixteen electronic fuses.

What it replaces. Imported zonal controllers and gateway chips that bridge older car networks to Ethernet. Why this process. A gateway is about how many channels it has and how predictable its timing is. It is not about clock speed. The network standard bounds the delay in the switch, not in the processor. Where it sells, and how many. We estimate 4 to 12 million units a year in Indian zonal and gateway sockets at \$6 to \$20 each. The commercial hook. A zonal layout replaces the old relay-and-wiring-harness box. That is a cost and weight saving the car maker argues for on your behalf. What it does not do. 130 nm at 200 MHz handles the local layer, not the central car computer. A central car computer needs multi-gigahertz processors, a graphics unit, an AI unit and fast memory. That is a much smaller process, and we do not claim it anywhere in this document.

9.10 Track B · D100 drone chip What it is. Flight control, plus position tracking from cameras and motion sensors, plus optional AI. It is built for transformer AI models, with the attention calculation in hardware using integer maths. We estimate that uses about 70 % less power than a general AI accelerator. It has lockstep RISC-V processors on a path to DO-254 aviation approval, and targets a chip cost under \$3.

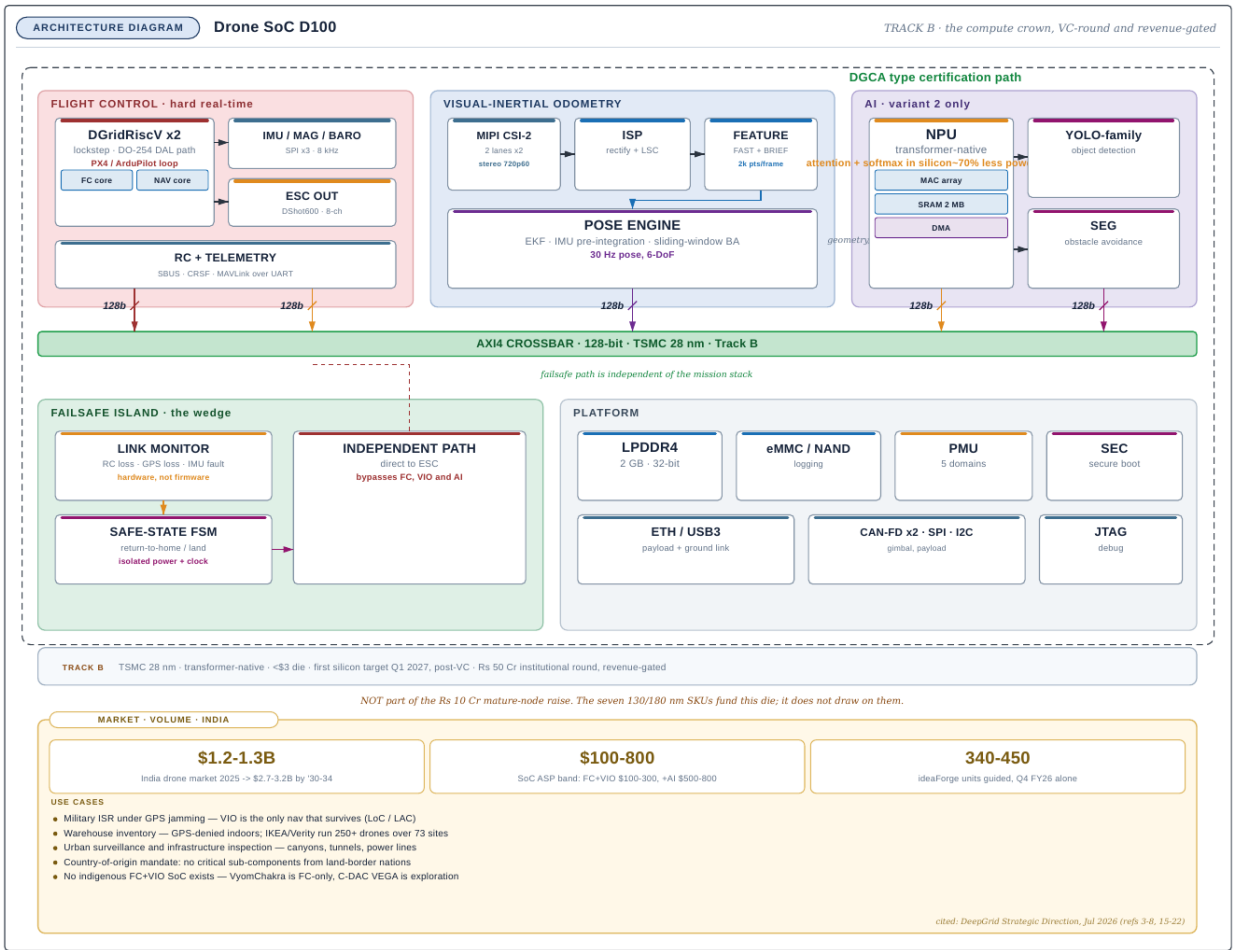


Figure 19. Drone chip D100 on TSMC 28 nm. Track B, the high-performance one, funded by a separate round and gated on revenue. Flight control, camera-based position tracking, optional AI, and a hardware-isolated safety section.

Anchor customer, named: Chakravayu CPDL, a drone maker — the first external buyer for the D100. Our own drone module flies it first. **PLANNED**

What it replaces. The Qualcomm QRB5165, which is the only real competitor and is imported. Everything else in this class does AI or flight control, not both. Where it sells, and how many. The Indian drone market was worth \$1.2 to \$1.3 billion in 2025, and is forecast at \$2.7 to \$3.2 billion by 2030–34. Flight control with position tracking sells at \$100 to \$300. Adding AI takes it to \$500 to \$800. ideaForge alone guided to 340–450 units in one quarter of FY26. The legal or demand hook. No Indian chip does flight control and position tracking together. VyomChakra does flight control only. C-DAC's VEGA is a research programme. And country-of-origin rules block components from countries bordering India outright. The money boundary, said twice because it matters. The drone chip is not part of the ₹10 Cr round. It is Track B: a separate ₹50 Cr institutional round, gated on revenue, with first silicon targeted for early 2027 after that round. The nine older-process chips fund it. It does not take money from them. How it connects to the other nine. The safety section is the real advantage. No chip company sells hardware-isolated return-to-home as a chip feature. And that block can be built at 130 nm. So we prove it on the older-process chips before committing it to a 28 nm mask set. Ten chips, each aimed at a socket an import fills today. But working silicon is not a product, and the distance between the two is where most hardware plans quietly fail.

10 From a working chip to something a customer can buy

Three crore of the ten-crore round goes here rather than into design. It is worth explaining why.

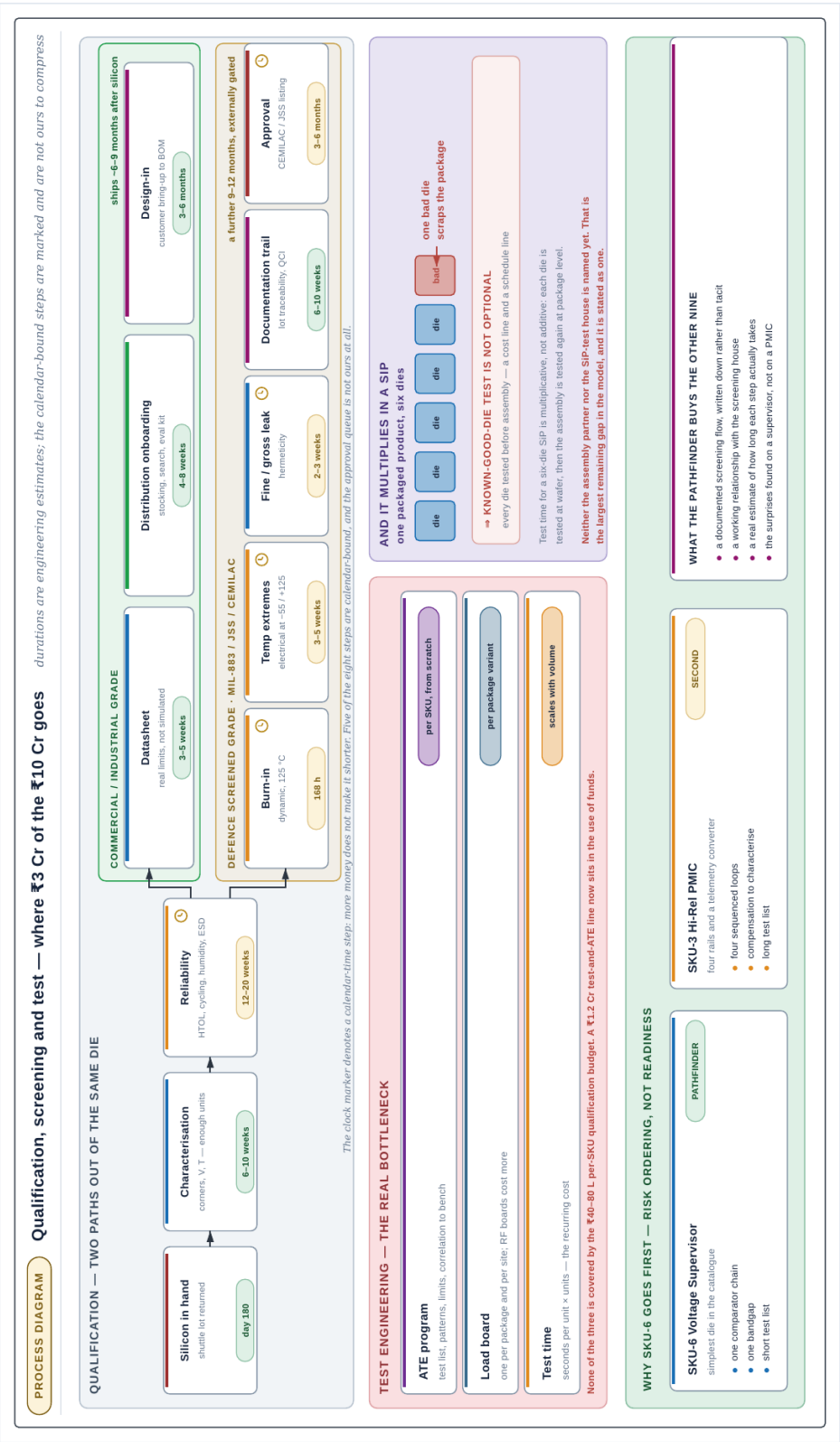


Figure 20. Two routes out of the same chip, with the calendar-limited steps marked with a clock. Five of the eight steps to a screened chip cannot be made shorter with money. The pink panel is the cost line that was missing from our own earlier model. The purple panel explains why putting six chips in one package forces you to test every chip first.

10.1 What ₹40–80 L per chip actually buys

Approval is not one test. For a commercial industrial chip it is a defined sequence with a defined number of samples. For a screened defence chip it is longer. Characterisation means measuring the chip across manufacturing variation, voltage and temperature, on enough units to write a datasheet with real limits instead of simulated ones. Reliability means high-temperature life testing, temperature cycling, humidity, static-discharge and latch-up testing. And for defence versions, screening means burn-in, electrical testing at temperature extremes, leak testing on the package seal, and the paperwork trail that MIL-883, JSS and CEMILAC approval require. The reliability and screening steps take calendar time. More money does not make them shorter. Five of the eight steps to a screened chip are in that category. This is why the plan sends commercial chips first, and pushes chip 6 through the process early — specifically to learn where it slows down before four other chips are queued behind it. One factory run cannot approve a chip, and that costs money. A shared run gives you one set of wafers, usually made at typical settings. An approval needs material from more than one batch, so that batch-to-batch variation is included in the samples. A defence screening process will ask for it. So ₹40–80 L per chip is honestly a commercial-grade approval figure. A screened version needs extra material, and for the analogue chips that means extra factory runs, not just extra test time. We have sized the round for four approvals rather than seven for exactly this reason, and §13 names the three that are not funded.

10.2 The cost line that was missing from our own model

TESTING WILL BECOME THE BOTTLENECK BEFORE DESIGN DOES A test program, a test circuit board and machine time are not covered by the approval budget. And for a package holding several chips, the test time multiplies rather than adds. In our earlier plans this line did not appear at all. A ₹1.2 Cr testing line is now listed separately in the use of funds. The assembly partner and the package-test company still need to be named, and that is the largest remaining gap in this document. We would rather you read it here than find it. There is a second consequence for the multi-chip package. If one chip in six is faulty, the whole package is scrap unless every chip was tested before assembly. So you must test every chip individually first. That is both a cost line and a schedule line, and neither existed in our model before this document.

10.3 The customer approves the supplier, not just the chip

This is the gap that an aggressive read of our own document found, and it is the one we had thought about least. Being on an approved-vendor list is not granted to a chip. It is granted to a company. And a defence or car customer audits the company before it audits the chip. They want a quality management system such as ISO 9001, or AS9100 for aerospace. They want records of approved suppliers. They want static-discharge controls. They want to trace every unit back to its wafer. They want a documented process for notifying customers of changes. And they want a second source, or a policy for last-time purchases. None of that is difficult, and none of it is expensive next to a mask set. But it takes calendar time, it needs somebody whose job it is, and it does not currently appear anywhere in our use of funds. A company with excellent silicon and no quality system does not get onto an approved list. The sales process in §11 has an approval step that quietly assumes this problem is solved. It is not. We would rather name it here than discover it in a customer audit. It is now in the risk table in §14. An approved chip is still not revenue. It becomes revenue through two completely different sales processes, both using the same chip.

PART FIVE

Two ways to sell the same chip, and why nobody can copy us The commercial route is self-service. The defence route is relationship-led and protected by law. One approved chip, two prices 10 to 100 times apart. And a big competitor with no available answer.

11 Two ways to sell the same chip

Selling chips is not a pitch. It is a sequence of things you hand the customer. A design engineer downloads a datasheet, decides in ten minutes whether the chip can work, buys an evaluation board with a credit card, gets it running in a week, and only then does anyone discuss commercial terms. If any item in that chain is missing, the process stops there, no matter how good the silicon is.

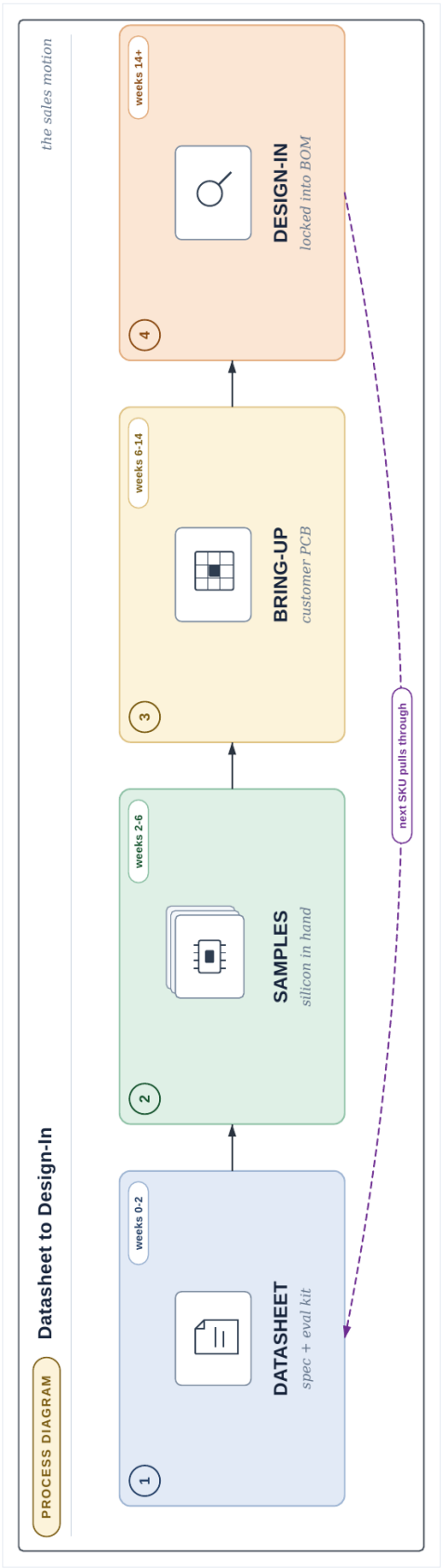


Figure 21. The commercial route as the drawing actually shows it: Datasheet in weeks 0–2. Samples in weeks 2–6. Getting it working on the customer's own board in weeks 6–14. Designed into their product from week 14. Each step hands the customer something they can act on without talking to us. And the dashed line back is the part that compounds: the next chip sells on the relationship the last one built.

That is why ₹1.0 Cr of the round is sales and working capital. And it is why the distributor conversation matters more than it looks. A catalogue chip that is not in stock and cannot be found in a search does not exist to the engineer who needs it. The defence route works differently. You find the item on SRIJAN or a PIL. You have a technical discussion with the defence company. You supply trial units. You get a Make-II or direct order. Then you go on the approved-vendor list. The trial stage is where a small company wins, because the buyer is risking time rather than money, and an Indian-designed part beats an import before price is discussed. Both routes use the same approved chip, at prices 10 to 100 times apart. That is the real efficiency of having a range, and it is why low volume is still good business here.

11.1 What we found when we read the government lists ourselves

We read the defence ministry's 5th PIL — 346 items, scanned, so we had to run text recognition on it — and DMA lists 1, 2, 3 and 5, directly rather than relying on summaries. What we found was not what we expected, and it changed how we sell. The lists name boxes, not chips. They list swappable units, assemblies and spare parts. Most of PIL-5 is bushes, valves, pumps, gaskets and optical filters. There is no chip-level item anywhere in what we read.

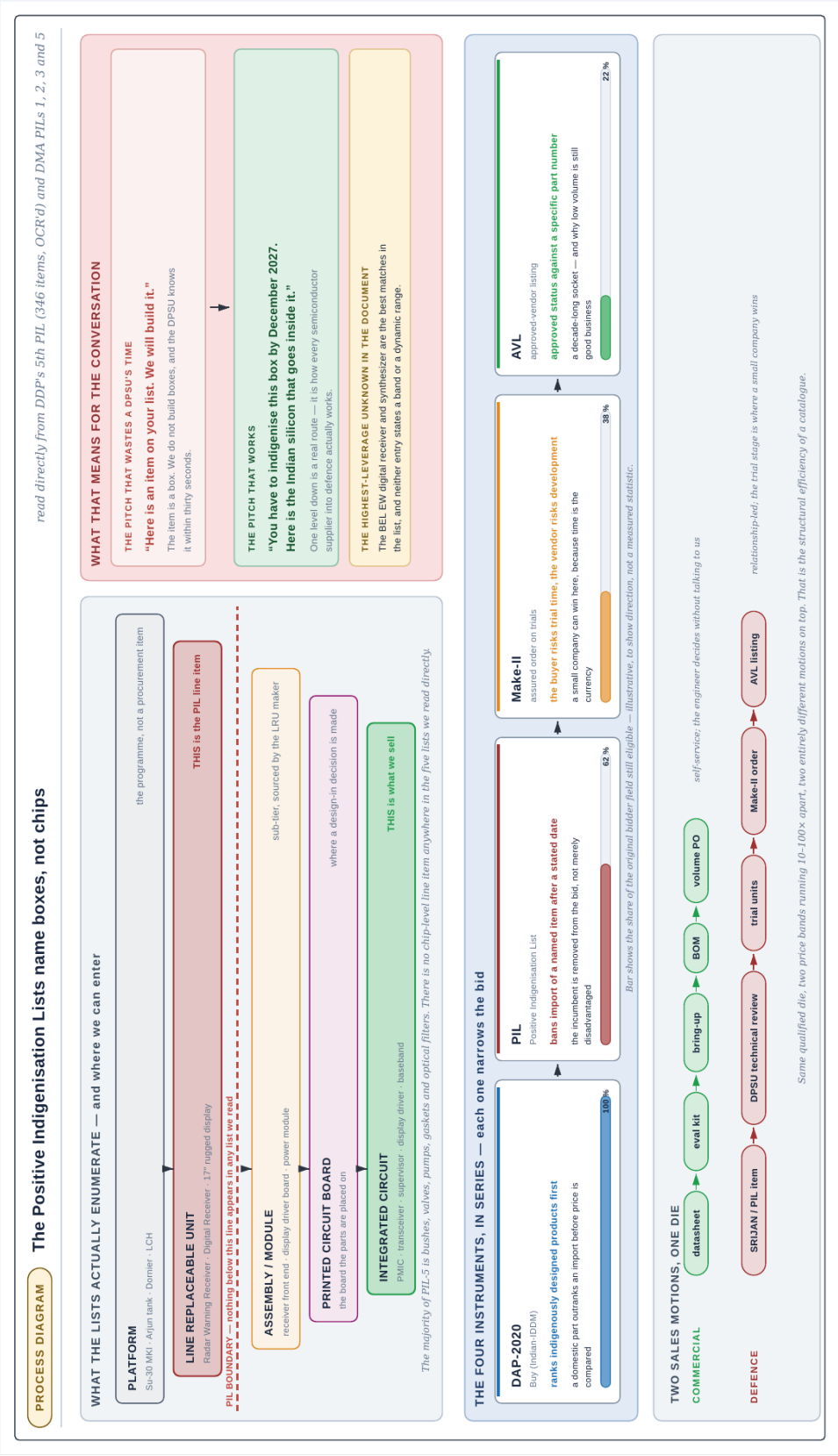


Figure 22. The finding, drawn. The lists stop at the swappable box. We sell three levels below that. The four legal rules then apply one after another, each narrowing the field of bidders. And the two sales routes both use the same approved chip.

So our entry point is one level down: supplying the silicon inside a listed box. That is a real route. It is how every chip supplier into defence actually works. But it is a completely different first meeting from “here is an item on your list, we will build it”. Pretending otherwise wastes a defence company’s time and our credibility. Electronics items we confirmed, that match our range: List Item Owner / platform Due Matches PIL-5 BLDC motor with encoder anti-tank missile Dec 2025 chip 1 PIL-5 DC-DC converter, 4 A, 16–40 V in BEL · tank Dec 2026 chip 3 PIL-5 DC-DC converter 12 V BEL · tank Dec 2028 chip 3 PIL-5 Isolation amplifier BEL · tank Dec 2028 nearby PIL-5 Digital receiver (SDC V2) BEL · electronic warfare Dec 2026 chip 7 family PIL-5 Synthesizer BEL · electronic warfare Dec 2026 chip 7 family PIL-5 17-inch rugged display BEL · Defence Electronics Dec 2027 chip 8 PIL-5 Radio altimeter HAL · Dornier Dec 2025 chip 7 variant PIL-5 Magnetometer HAL · LCH Dec 2025 nearby DMA Radar warning receiver Su-30 MKI, Mi-17 Dec 2025 chip 7 family DMA Weapon locating / battlefield radar Army 2022–26 chip 7 family DMA Power amplifiers, 10 W and 2–4 kW Air Force Dec 2021 IHP SiGe DMA Software defined radio, manpack all services Dec 2022 future DMA IRNSS receiver; GNSS jammer; GNSS spoofing Navy / Air Force Dec 2022 future DMA Proximity fuze, 76/62 SRGM Navy — future The highest-value match is also the least specified. The BEL electronic-warfare digital receiver and synthesizer are the best matches in the whole list. And the one-line entries give neither the frequency band nor the dynamic range. That single answer decides whether we use the SiGe process or ordinary CMOS. Which then decides the cost, the schedule, and whether it is one chip or two. It needs a conversation with BEL, not more desk research. It is action 4 of seven in §14. Two routes, one chip, and a set of laws that removes the competitor from one of them. The obvious question is what that competitor does about it.

11.2 The rule that guards the money: no buyer, no build

A chip gets build money only when its buyer has a name — our own product needs it, a government make-in-India list names it, a defence filing names it, a tender asks for it, or a customer signs a letter. Nothing else counts. Not enthusiasm, not a market report, not a deadline.

Today the scoreboard reads: **eleven** chips have a named buyer and may spend — 1 (anchor: Airgap Technology), 3, 4 (anchor: MCEME, contracted), 5, 6, 7, 8 (buyer and date on a government list: BEL, Dec 2027), 10 (anchor: Chakravayu CPDL), plus the security chip and the drone-navigation helper our own products need. **Four** wait on one signature — chip 2 (Ripple’s letter; our biggest revenue line, gated first on purpose so the rule is seen to be real), chip 9 (a vehicle maker’s pilot), the cockpit display controller (rides chip 8’s buyer), and the avionics power part (opens when chip 3 passes screening). **Eight** are on hold until a programme, mandate or letter names them — and spend nothing meanwhile. The newest two are the V2X vehicle-communications set (SKUs 21–22): a 5.9 GHz radio front-end on IHP’s radio-grade 130 nm process, and an LTE-class road-safety modem on the same fab’s digital side, carrying the message-signing security engine built from chip 15’s block and leaning on chips 4 or 9 as host. The bound is stated: the newer 5G flavour of the standard is a finer-node problem, and we do not claim it. India has no V2X mandate yet, so both wait — a chip on hold is the rule working, not a gap.

Three kinds of buyers run on three clocks, and mixing them up is how forecasts lie:

- **A list entry — permission, not an order.** The import ban arrives on the list’s date; orders trickle in over 2–4 years. Good for the future; never counted in this year’s number.
- **A defence filing — the real order path.** File, build a sample at our cost, pass trials — about 15–18 months — and then the order is assured by rule. The only path we count revenue on.
- **A tender win — the fast clock.** Get written into a tender’s requirements and ship on its schedule. Meters and fans live here. Hidden step: just registering as a vendor takes 6–12 months — it is on our timeline now.

11.3 Who actually buys — names, numbers, dates (September 2026)

Facts as of September 2026, sources on file. Labelled so nobody mistakes the size of a door for an order of ours.

- **Smart meters.** 20.33 crore meters sanctioned nationally; 7.24 crore installed by June 2026; tenders for ~5 crore meters running now in Haryana, MP, Punjab and Tamil Nadu; the scheme runs to March 2028. The two biggest doors: Genus (order book ₹24,020 crore; an 18-million-meter yearly factory) and Adani Energy (over 4.7 crore meters in hand). *Their orders, not ours — this shows the size of the door.* Our anchor, Ripple Metering, is the named first door.
- **Defence.** A sixth import-ban list arrived in August 2026: 405 items, about ₹3,070 crore of business. Over 33,000 items sit on the SRIJAN portal for indigenisation, and defence companies have already placed ₹7,572 crore of orders on Indian vendors. BEL runs the display line our chip 8 answers.
- **Drones.** The Army now physically tears drones apart to verify where every part comes from; recent orders (₹137 crore, then ₹105 crore, to ideaForge) went only to platforms that passed. An Indian-made drone brain is exactly what that inspector wants to find. ₹40,000 crore sits in the new emergency-procurement cycle. Our anchor here: Chakravayu CPDL.
- **Electric vehicles.** 14 lakh electric two-wheelers sold in FY26, up 22%, crossing 2 lakh in July 2026 alone. Five controller makers to design into: TVS, Bajaj, Ather, Hero Vida, Ola. Our anchor, Airgap Technology, builds the motors these controllers drive.

12 Why a big competitor cannot copy this

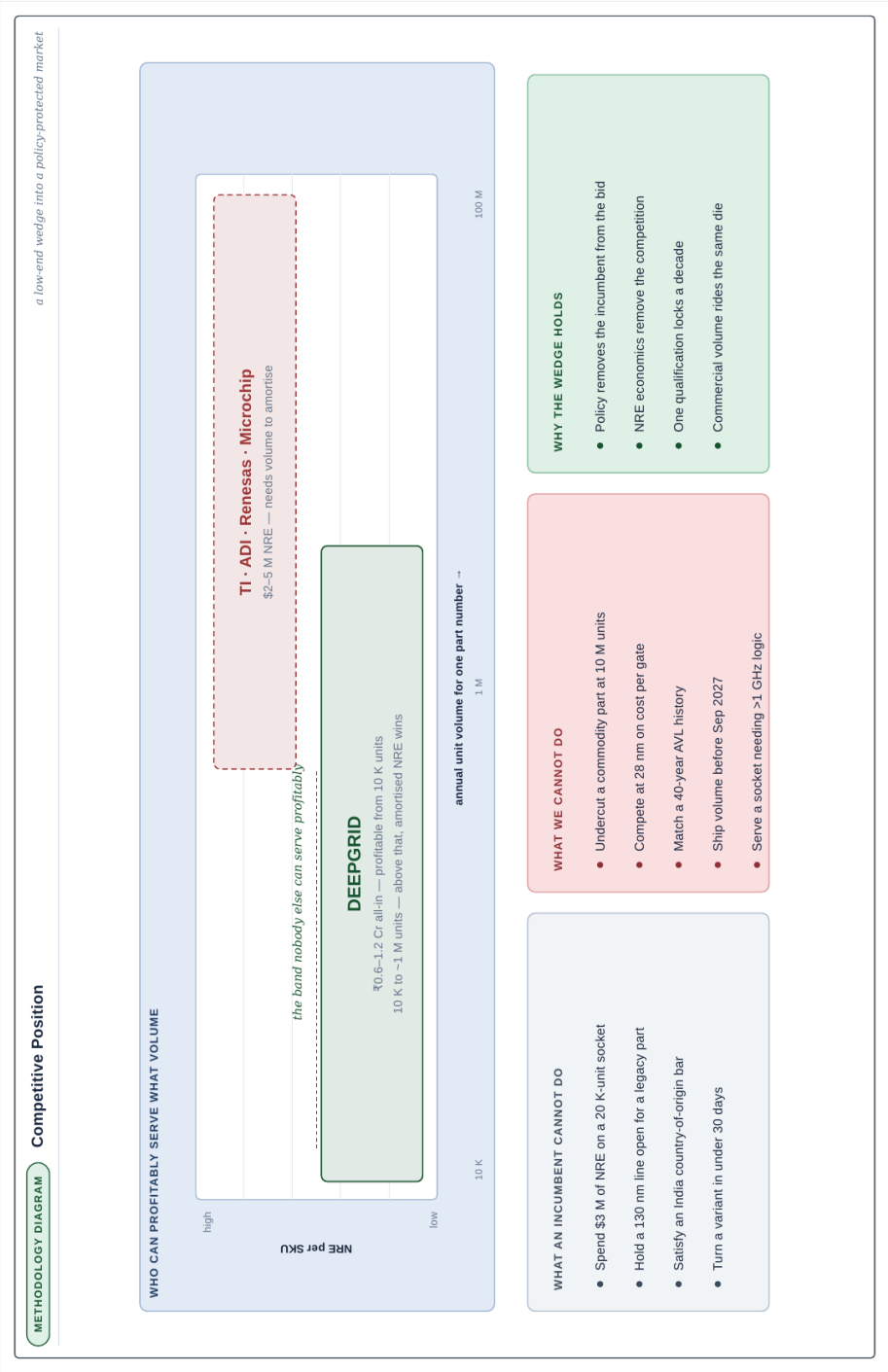


Figure 23. Where we sit. The two axes are the only ones that matter here: setup cost per chip, against how many units of one part number sell per year. The green area is 10,000 to about 1 million units. Above that, a big company's paid-off setup cost and cheaper wafers win, and the panel headed "what we cannot do" says so in its first line.

In the commercial market we are against Texas Instruments, Analog Devices, Renesas, Microchip and ST. They win on range, distribution, software tools and trust. We do not claim otherwise. What we do is serve a shape of demand their costs exclude: a 10,000-unit programme with a specific requirement and a screening step. In the defence market the competitor is not a chip company winning on merit. It is an import that the law is progressively removing from the bidding. Our real competitor there is another Indian company, and what separates us is who has finished silicon and a screening process first. Today, we do. In the drone market it is Qualcomm's QRB5165 and nothing else integrated. That is a Track B argument, funded separately. A big competitor has three possible answers, and all three are blocked. Cut the price does not help, because the barrier is paying off the setup cost on a small programme, not the unit price. And after a PIL date the price is irrelevant, because the chip cannot be chosen at all. Open a low-cost design centre in India is real, and several have. It reduces their engineering cost but not their \$2–5 million per-chip layout and mask cost. And it does not make the chip Indian-designed for DAP-2020 purposes. Buy the new company is the likely ending, and it is an acceptable outcome for this round. But it is not a plan we depend on.

12.1 How this strategy fails, said plainly

Companies that enter at the low end often get stuck there. The pattern is well documented. The new company becomes dependent on its protected corner, never builds the ability to compete on merit, and gets squeezed when the protection ends. It would be dishonest to present the entry without the failure mode. Our answer is the block library and the process roadmap. What compounds is not the protected socket. It is the library of silicon-tested blocks, and the method that produced it. Both can move to processes and markets where nobody is protecting us. The move to 90, 55 and 45 nm from 2028 exists for exactly this reason, and it is driven by channel count and signal processing throughput, not by clock speed. That is the whole argument. What is left is the money: what it buys, what it proves, and what could stop it.

12.2 What if Chinese prices crash — and our stop rules

The cure for loving your own forecast is a second forecast where things go badly. Here is FY31 re-priced as if Chinese chips flood India — and note which row the crash cannot touch:

Row	Our plan (₹Cr)	If prices crash	Why
Smart meters	480	~340	tenders squeeze prices: –30%
Motor controllers	220	~130	the most exposed chip: –40%
Defence (screened)	200	200	import-banned market — the crash cannot reach it
Drone brain	65	~55	Army origin checks blunt the war
Vehicle gateway	35	~25	open market: –30%
Total	1,000	~750	even then the plan survives, and the next round still triggers

And our stop rules — written now, with dates, so nobody bends them later in an excited meeting:

- **S1.** If Ripple (or another meter maker) has not signed by the day chip 2's factory order is due, that chip waits one full cycle and its money moves to chips 1 and 3. No exceptions, however the meeting feels.
- **S2.** If chip 6 fails military screening twice, every document moves defence revenue out by twelve months, within thirty days.

- **S3.** In FY29 we check Chinese prices properly — duties and tender rules included. If they land below what the chip costs us to make, we leave the fan market and keep e-scooters plus our own products. The walk-away price gets calculated this quarter, calmly.
- **S4.** If the government factory in Mohali runs more than two cycles late, we say so publicly and lean on our two other factories. No quiet slipping of dates.

One piece of homework we owe ourselves this quarter: find out, properly, what Chinese chips will really cost in India in FY29 — duties and tender rules included — before an investor asks us. And one line for §12.1's list: *the meter row fails loudest — if no letter is signed by the factory-order date, that ₹480 crore moves out a year, and the plan still closes near ₹520 crore on the other rows.*

13 What ₹10 Cr buys, and what it proves

Under contract today: ₹1.01 Cr. Live including confirmed commercial orders: ₹2.88 Cr. Everything beyond that is a scenario, and we present it as one. Milestone When What it means All 6 prototypes made and tested Mar 2027 parallel factory runs, one per cycle Commercial production Sep 2027 motor controller and meter chip ship first FY27 · ₹4 Cr FY27 ₹2.88 Cr of it already live, before any chip FY28 · ₹28 Cr FY28 commercial volume ships; defence batches enter screening FY29 · ₹140 Cr FY29 3–4 more chips; meter volume ramps FY30 · ₹410 Cr FY30 approved-vendor listings build up; defence at screened prices FY31 · ₹1,000 Cr FY31 meters ₹480 Cr · motors ₹220 Cr · defence ₹200 Cr · drone-related ₹100 Cr

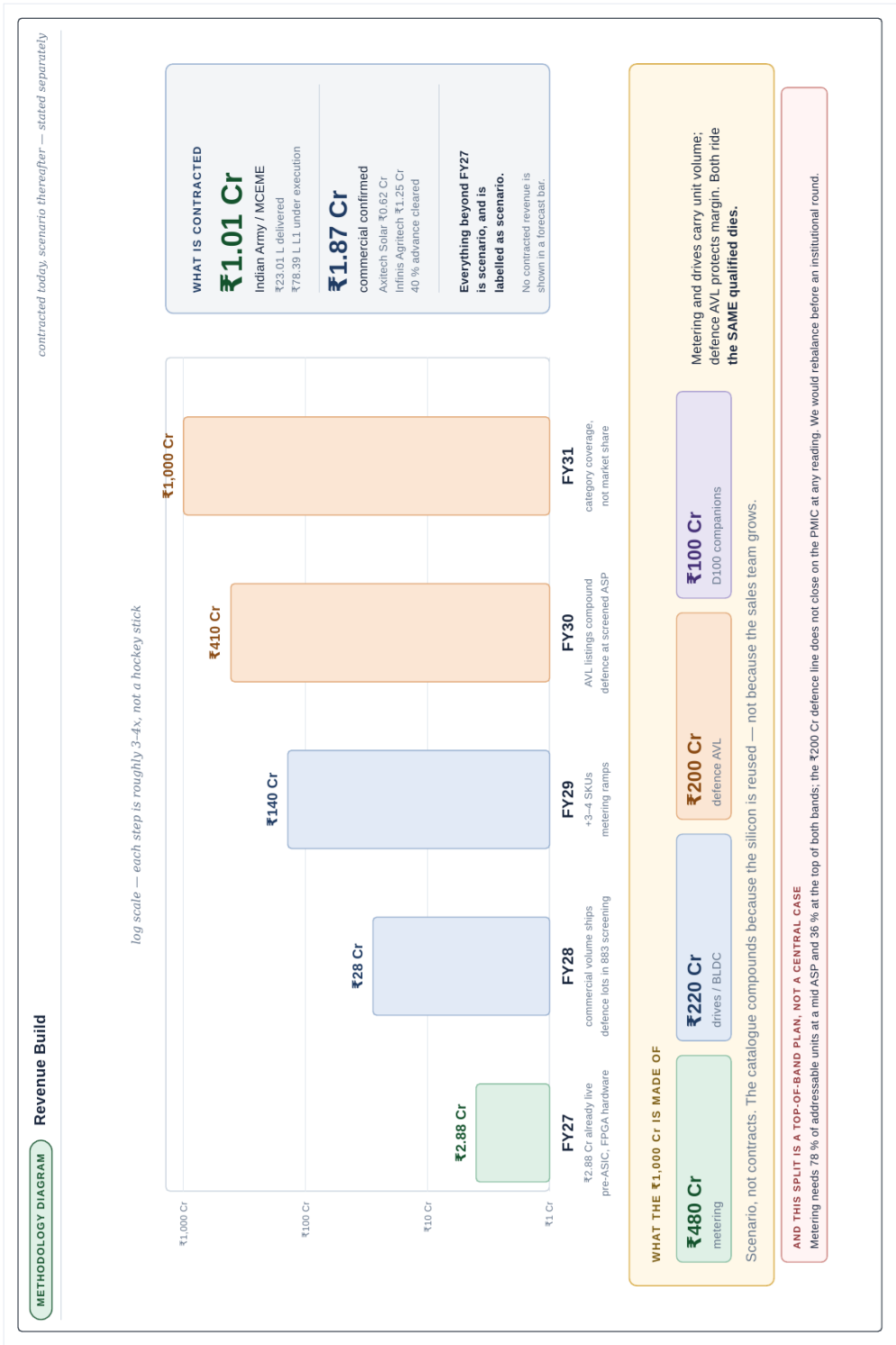


Figure 24. Revenue build-up. Contracted revenue and scenario revenue are drawn in clearly different colours, because mixing them is the fastest way to lose a careful reader.

The FY31 figure is coverage of categories, not market share. By year 5 our range covers most of the older process chip categories India imports. It only works because the meter and motor volume uses the same approved chips the defence versions use. One approval, two prices, two markets. We then ran the obvious check on our own number, and it did not fully pass. WE CHECKED OUR OWN FY31 SPLIT, AND PART OF IT DOES NOT HOLD Take the FY31 figure, divide it between the nine chips, then divide each chip's share by its price. That gives the number of units each line needs, and therefore what share of the market it needs. At ₹95.39 per dollar, the meter chip needs 72 % of our stated volume band and the motor chip 42 %, at mid prices. At the top of both bands those fall to 34 % and 19 %. If we cap every chip at a 30 % share, the nine chips can still carry ₹1,324 Cr. So the total holds and the split does not — 70 % of FY31 sits on two chips, while chips 9, 7 and 4 are planned at under a tenth of theirs. We would rebalance before an institutional round. One caution on how to read those percentages. They treat our stated volume band as the whole market. An investor's own sizing model reads the same bands as our target inside a larger category, and quotes separate category sizes for five of the ten chips. On that reading the shares are 11–30 % of the category, not 42–72 % of the market. That is probably the better reading, and it would remove the alarming figures above. We have not yet rebuilt the table that way because a category size exists for only five chips. What survives either reading is the concentration, which is why the rebalancing point stands.

13.1 What the money physically produces

The first question a hardware investor asks is not about milestones. It is about units. So: Prototypes 600 chips six factory runs $\times$ 100 packaged parts. Not sellable — not approved, and 0.6 % of one year's need Production 94,000–165,000 production mask sets and first wafer batches for chips

1 and 2 only

Value of those batches ₹2–9 Cr at the \$3–8 and \$2–5 prices already stated FY27 needs 100,000 units at an average \$4.50 — the two first batches cover FY27 completely FY28 needs 707,000 units about eleven more batches per chip, which are working capital against orders, not setup cost, and not in this round The arithmetic: a 200 mm wafer, 3 mm unusable at the edge, 90 % of the rest usable after cutting lanes, 85 % of chips working, 25 wafers per batch. The chip sizes are engineering estimates — 8 to 15 mm² for chip

1 with its high-voltage section, 6 to 10 mm² for chip 2. That is the sensitive input. If chip 1 comes out at

15 mm² rather than 8, the answer drops to the bottom of the range and FY27 coverage gets tight. We will

publish measured sizes with the first layout. This is the whole case for funding a production batch inside a seed round. A prototype chip costs about ₹14,300. A production chip costs ₹120 to ₹260 in mask cost spread across the batch. There is no volume at which buying more prototype runs beats buying one mask set. The prototype run proves the chip. Only the mask set can sell it. A round that funded prototypes and stopped would produce 600 chips, no revenue, and a company that must raise again before it can invoice anyone.

13.2 The tightest link, now that we have the factory dates

Six designs split across the CI2609 and CI2612 runs do get sent for manufacture before March 2027, so that milestone holds. But chips arrive on 3 March 2027 for the first three and 25 May 2027 for the second

three. That leaves roughly four months to characterise, release and reach production by September 2027. That is the tightest link in this plan. The fix is a sequencing decision, not money: put chips 1 and 2 — the two the revenue plan depends on — on the CI2609 run, and let the later run carry the chips whose revenue is FY29 and beyond.

13.3 Money, time and the size of the team

An investor's second question, after whether the technology is real, is how long the money lasts and how many people are spending it. This document did not answer either until we were asked. So: Runway on ₹10 Cr

24 months, to the March 2027 milestone and into first production

Average monthly spend about ₹41 L, weighted early to engineering and late to masks and approval. Engineers funded 5–7 people on ₹2.4 Cr over 24 months — ₹1.2 Cr a year at Hyderabad costs. What that team is

1 senior, 3 mid-level, 2 junior, of whom 2 are analogue designers

Founder pay partly deferred, which is what makes 5–7 fit inside ₹1.2 Cr a year. Revenue during the period ₹4 Cr in FY27 and ₹28 Cr in FY28, of which ₹2.88 Cr is already live. And the honest problem in those numbers. Five to seven engineers, only two of them analogue, against four analogue-heavy chips each needing two or three silicon attempts, is thin. We ran that check ourselves at ₹1.2 Cr a year of fully loaded cost and it does not support a bigger team. So we would rather show the small number than quote a bigger one the budget cannot buy. That is the real reason the round funds four approvals and not seven, and why chips 5, 7 and 8 are scheduled into the second cycle and later, rather than promised in this window.

13.4 The ask

₹10 Cr angel or seed, on a simple convertible instrument, capped below the institutional round. It funds six parallel factory runs, four approvals, and working capital. That means seven older-process chips from architecture through tested silicon and into approval, including production mask sets and first wafer batches for chips 1 and 2. Nothing is drawn for the 28 nm drone chip, which is Track B and gated on revenue at a ₹50 Cr institutional round.

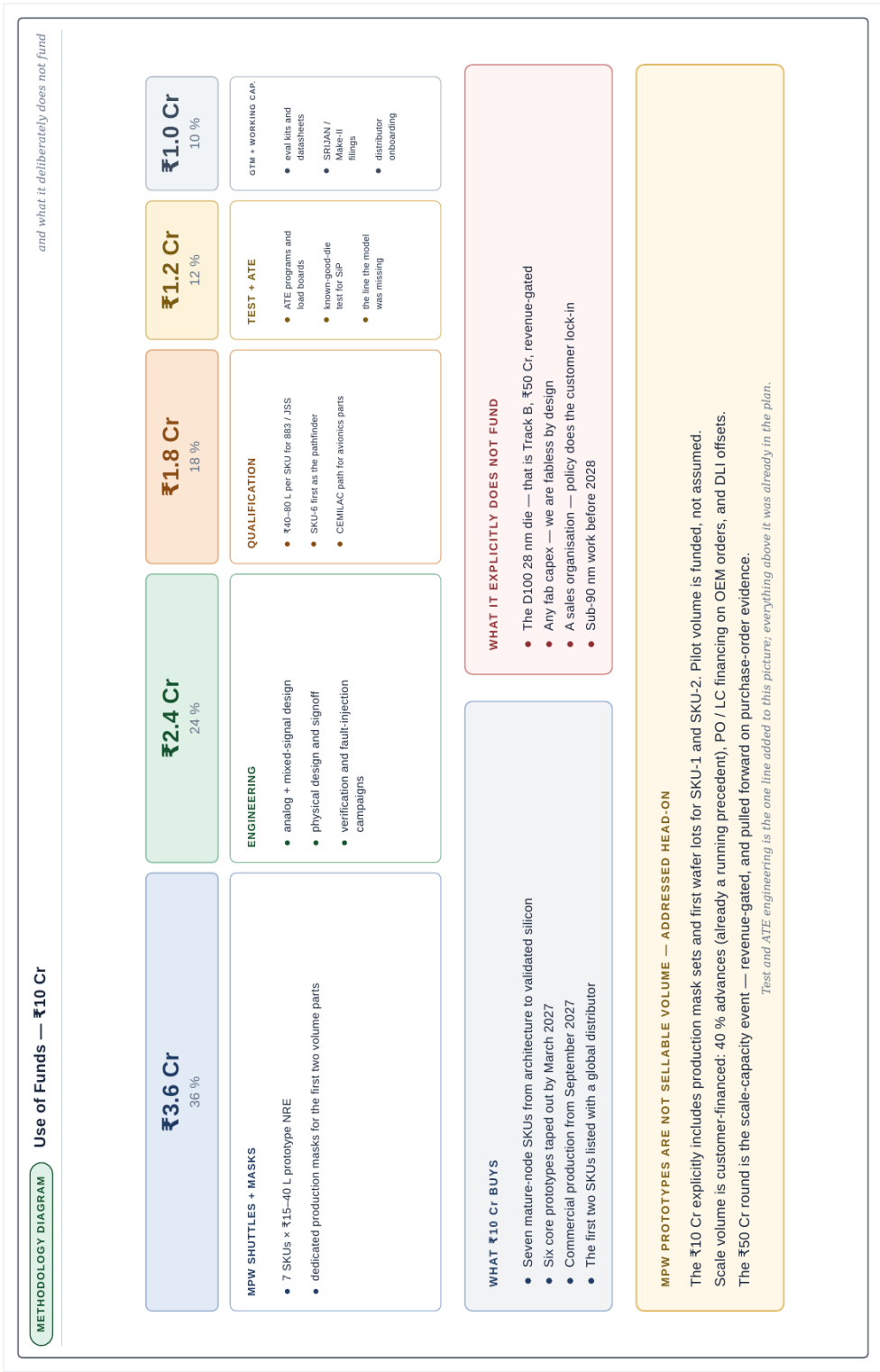


Figure 25. Use of the ₹10 Cr, and on the right, what it deliberately does not fund. The testing line is new in this version of the model, for the reason given in §10.

Allocation Amount What it buys Factory runs and masks ₹3.6 Cr ₹0.86 Cr for six sky130 runs plus ₹0.48 Cr for the IHP SiGe chip — ₹1.34 Cr of prototypes — leaving ₹2.26 Cr for two production mask sets and first wafer batches Engineering ₹2.4 Cr design, verification and layout staff Approval ₹1.8 Cr four approvals including one MIL-883 first-through Testing ₹1.2 Cr test programs, test boards, machine time Sales and working capital ₹1.0 Cr datasheets, evaluation kits, distributor setup, stock The round is not sized to reach the FY31 number, and it should not be. It is sized to reach the point where this stops being a design story. Seven chips in silicon. Four of them approved. Two in volume production. And a revenue line that is measured rather than modelled. That is the state where an institutional round is priced on evidence instead of on a story. Getting there is the entire purpose of this ₹10 Cr. Which leaves the only honest way to end a document like this: what could stop it, and what we would fix first.

13.5 What each chip must earn — the one table

Our funnel ends at ₹1,000 crore by FY31 and calls it an ambitious plan. Fair. This table breaks that one number into ten small ones, so a reader can argue with any single row instead of taking the total on faith. Every cell is **ESTIMATED** unless tagged otherwise, and the FY31 column adds to exactly ₹1,000 crore: 220 + 480 + 200 + 100.

Chip	Sells for	FY29 → ₹Cr	FY31 → ₹Cr	Margin	How much of the market this needs — honestly
1 Motor controller	\$3–8	~1.0M → 40	~5.5M → 220	30–40%	about 1 in 3 such chips sold in India — ambitious, and open to Chinese price attack
2 Smart-meter chip	\$2–5	~2.4M → 60	~19M → 480	35–45%	at mid price, 78 of every 100 meters — i.e. winning 2–3 big meter makers outright
3 Power chip (28 V)	\$50–200 scr.	~8K → 8	~45K → 55	60–75%	a small slice — but only <i>after</i> military screening
4 Safety processor	\$4–12	~60K → 5	~0.35M → 40	45–65%	under 1 in 10 of India's safety-chip sales
5 Line driver	\$0.5–1	~0.2M → 3	~1.4M → 22	40–70%	rides along wherever chips 3 and 4 sell
6 Voltage watchdog	\$0.2–0.8	~0.5M → 2	~3M → 13	40–70%	on almost every board; small money on purpose
7 Radar front end	\$35–60/set	~5K → 4	~75K → 30	50–65%	our own radar product first; chips in FY28
8 Display driver	\$70–110	~6K → 5	~55K → 40	55–70%	a government list names the buyer <i>and</i> the date
9 Vehicle gateway	\$6–20	~50K → 3	~0.7M → 35	35–50%	the simpler vehicle box only — we do not claim the big computer
10 Drone brain (D100)	\$60–100	~15K → 10	~110K → 65	45–55%	anchor named: Chakravayu CPDL; our own module flies it first

One-time cost per chip stays as \$10 states it: ₹0.6–1.2 crore all-in. The radar is two chips — its radio part can only be proven in real silicon — so budget ₹1.4–1.8 crore there. Every chip except 5 and 6 earns back its one-time cost within its first year of sales; those two earn theirs by riding along with everything else, and chip 6 carries the military screening that opens the defence door for the whole family.

The bottleneck nobody plans for: *fifty chips by Year 5 means three hundred customers being helped at once, because each chip needs five to ten customers and each customer needs months of our engineers' help. The factory is not the limit — support engineers are. We measured, on our real kit customers, that one engineer can handle 25–35 of them, and the next funding round pays for ten to twelve support engineers as a proper line item.*

14 What could stop this, and what we would fix first

Risk Why it is real What we do about it 200 MHz is a design target, not measured silicon we have not published the timing analysis, and it is unclear which factory kit it was run against run it on sky130 and publish the report before the number goes to a customer One mistake costs a whole cycle the schedule is locked to factory dates; about 6 months per miss enforce design-complete before the factory window; use parallel runs so one slip does not stop everything Testing becomes the bottleneck ₹40–80 L per chip covers approval, not test programs, test boards or machine time ₹1.2 Cr testing line now listed separately No packaging supply chain yet a six-chip package needs an assembly partner, package testing and per-chip screening first name the assembly partner; add per-chip testing to the model One supplier now covers design to volume the same overseas service would handle both the prototype and the production run; their published position is prototypes only, and the production route is a verbal indication, not a quote get the written production quote (action 3); SkyWater direct is the commercial backup, SCL the Indian one No quality management system yet an approved-vendor listing goes to a company, not a chip; the customer audits ISO 9001 or AS9100, supplier records, static-discharge control and traceability first name an owner and start it alongside the first factory run; it costs time, not capital A free design kit has no tested analogue library the ± 15 kV protection ring, the temperature-stable reference and the memory are targets with no example to copy budget analogue blocks as their own line and prove each one in silicon before it becomes a datasheet number Buy(Indian), not Buy(Indian-IDDM), until SCL IDDM depends on Indian value content, and the wafer is foreign until SCL production is real keep SCL as a business need, not a wish; do not claim IDDM priority before it is earned Analogue needs 2–3 cycles, not one four of the ten chips cannot be checked in silicon in a single cycle schedule those four across cycles 2 and 3; do not present one speed for a range with two The electronic-warfare opportunity is unspecified frequency band and dynamic range are unknown for the two best matches the BEL conversation, prioritised The radar front end cannot be tested on an FPGA

77 GHz cannot be prototyped on an FPGA at all

budget a SiGe test chip; do not schedule chip 7 as if FPGA testing reduced its risk The approval clock is not ours 9–12 months of MIL-883, JSS and CEMILAC cannot be shortened by us commercial chips first; chip 6 through the process early to learn where it slows

Risk Why it is real What we do about it Revenue is concentrated the FY31 split puts 70 % on two chips at top-of-band share rebalance toward chips 9, 7 and 4 before the institutional round No longevity promise published defence platforms run 20–30 years; a supplier with no last-time-buy policy is a bigger discontinuation risk than an import with a known end date publish a longevity policy on the first datasheet — at 130 nm it costs us almost nothing to honour We are the only source for every chip a critical part in a defence platform cannot have one supplier hold tested chip stock against customer forecasts, and qualify a second factory once SCL is real; neither is funded in this round PIL dates can slip some dates in the table above have already passed do not bet on one PIL date; the range covers four different markets Too few key people the method, the circuit code and the customer relationships sit in a very small team the method is scripted and reproducible from the source files; written down rather than in people's heads Of all of those, two are genuinely unresolved rather than managed: the 200 MHz timing analysis and the absence of a named packaging and testing partner. Everything else on the list has a fix we control.

14.1 The seven things we would fix first, in order of value

1. Publish what the 130 nm test chip actually cost, and what was on it. This is the correction we least wanted and most needed. The whole cost argument in §2 rests on a ₹0.6–1.2 Cr figure that has never been done once. Except that we have already made a chip on this method, so a real number exists and we are simply not showing it. It either supports the claim or it does not, and either way a reader is entitled to it before being asked to trust the estimate. It also answers the question behind it: did that chip contain any analogue?
2. Run the timing analysis on sky130 and publish it, per chip. The 200 MHz number is the biggest claim we make and it is currently a design target. One report per chip turns it into a fact, and it is the cheapest credibility available to us.
3. Get the production price quote in writing. Our factory service has said it is willing to support a direct production route. Their published FAQ still says prototypes only. A quoted mask set and first wafer batch turns the whole production half of this plan from a verbal indication into a number. That is the difference between a company that can invoice in FY27 and one that cannot.
4. Get the BEL electronic-warfare specification. It unlocks the highest-value matches on the government lists, and settles the SiGe-versus-CMOS question that gates the whole chip 7 family.
5. Replace target numbers with measured numbers. When the first 130 nm batch comes back, swap in real coverage, timing, yield and test results for the current targets. Every drawing in this document is built to accept that swap without being redrawn. That one change turns a plan into a track record.
6. Name the packaging and testing partners. It is the largest unaddressed gap in the model, and the easiest one for a technical reviewer to find.
7. Publish the TinyTapeout 6 project link. A public shared-die manufacture that anyone can verify is cheap credibility, and right now we are asserting it rather than showing it. None of the seven needs this round to happen, and five of them we can do in the next sixty days. We list them because a document that only argues its strengths is not a document you can act on. And because the first two came out of arguing against ourselves, not out of a reviewer's comments. WHAT WE ARE ACTUALLY ASKING YOU TO BELIEVE Not that we can out-engineer Texas Instruments. We cannot, and this document does not claim to. Not that the FY31 number is a forecast. It is a scenario, and we have just shown you which part of it does not hold.

What we are asking you to believe is smaller and easier to check than either. That an older process, a free design kit and an automated method take about ten times out of the cost of a finished lowvolume chip. That we have already made silicon using that method. That Indian law has created a buyer who cannot use the foreign supplier. And that ₹10 Cr is enough to put seven such chips into approved silicon. We are also asking you to accept a specific unfinished shape. The technology risk is mostly retired — there is silicon. The production risk is not. Nobody, including us, has taken a design made on a free kit through a full chip run and a dedicated mask set to an approved, listed part. That step is what this round buys, and that is the honest description of what ₹10 Cr is at risk against. Every number behind those claims traces back to something you can look at: a timing report, a utilisation report, a signed purchase order, a scanned government list, or a generated circuit code file. Pick any one and we will show you the thing behind it today, not after the round closes.

14.2 Where we could fool ourselves — the audit we ran on our own plan

We checked this plan for wishful thinking using investor Charlie Munger's famous checklist — the way we would tear apart somebody else's pitch. The core survives: the ten-times cost cut is arithmetic; ₹2.88 crore of orders is fact; the import bans are real. Four kinds of wishful thinking had crept in, and each one changed this document:

- **Borrowed glory.** We quoted other companies' huge order books in ways that made them sound like our orders. Fixed — they are labelled as the size of the door, and our money moves only on our own signed letters.
- **Permission mistaken for orders.** A list entry is permission, not an order. Fixed — the three buyer speeds in §11.2, each with its own clock.
- **Protecting our own number.** Our forecast added to exactly ₹1,000 crore in every revision — a sign we were protecting the number. Fixed — the crash-price table now travels with it everywhere.
- **One skill mistaken for another.** Being good at chips is not being good at supporting three hundred customers. Fixed — the support-engineer team is a funded line.

When everything pushes one way: *national pride, government deadlines, finished documents and exciting tender numbers all push the same direction — build more, faster. When every force in the room points one way, that is itself the warning. Our defence is the rule, the scoreboard, and the stop rules — written calmly, in advance.*

15 What happens next

Because a document should end with a step, not a summary. 1. A ninety-minute technical session, with the files open. Bring an engineer. The fastest way through this document is to pick the claim you trust least and start there. 2. A data room, which is ready and does not wait on the round: the sky130 manufacturing files and the TinyTapeout 6 project, the generated processor circuit code, the FPGA timing and utilisation reports, the three signed customer contracts, the scanned government lists, and the source files that regenerate every drawing in this document. 3. The instrument, deliberately boring: a simple convertible, ₹10 Cr, capped below the institutional round, with the drone chip ring-fenced. 4. Two things we will send you without being asked, within sixty days, whether or not this progresses: the all-in cost of the 130 nm test chip, and the per-chip timing report. Those are the first two of the seven actions above. We would rather you judge us on whether they arrive than on whether we promised them. Aravind Prasad · Managing Director · aravind@deepgridsemi.com · +91 74163 19944

16 Word list

Terms in the order a reader is likely to meet them. Term What it means 130 nm How small the smallest feature on the chip is. 130 nanometres is an older, larger, cheaper process. Newer phone chips are 3–5 nm. PDK, design kit The file set a factory gives you so you can design a chip for its process. Normally paid and under a non-disclosure agreement. sky130 and IHP's are free. RTL, circuit code The code that describes what the chip does, before it becomes a physical layout. GDSII The final file the factory needs. It describes every shape on every layer of the chip. Physical design, layout Turning circuit code into actual shapes and wires. Normally sent to an outside company at \$1–2 M per chip. Mask set The set of photographic stencils used to print the chip. A full set costs \$0.5–1 M.

Term What it means MPW, shared factory run Several companies' designs share one set of masks

and one wafer to split the cost. Cheap, but you only get about 100 chips. Wafer, wafer batch A 200 mm disc of silicon holding thousands of chips. A batch is 25 wafers. Yield The share of chips on a wafer that work. We assume 85 %. Die One individual chip, before it is packaged. QFN A common small chip package with pads underneath instead of legs. STA, timing analysis The check that proves a chip will work at its stated clock speed. FPGA A chip you can reprogram. Used to prove a design works before making a real chip. Slower than the real chip, which is why 81.25 MHz is not the product number. ASP Average selling price per chip. NRE Non-recurring engineering. The one-off setup cost before the first unit ships. AVL Approved-vendor list. Being on it means a buyer is allowed to purchase from you. PIL Positive Indigenisation List. A government list of items that may not be imported after a stated date. SRIJAN The government portal listing imported items that defence companies want replaced. DAP-2020 The defence procurement rules. Buy(Indian-IDDM) is the top priority category. Make-II A route where a successful trial earns a guaranteed order. MIL-883, JSS, CEMILAC Military testing and approval standards. Together they add 9–12 months. LRU Line-replaceable unit. A swappable box inside a vehicle or aircraft. DPSU Defence public-sector undertaking. A government-owned defence company, such as BEL or HAL. Analogue Circuits handling continuously varying signals, not just ones and zeros. Harder, slower to design, and needs silicon to verify. LDMOS, thick oxide Transistor types that survive high voltage. Available on older processes, not on newer ones. SiGe Silicon-germanium. A material that makes much faster transistors, needed for 77 GHz radar. DICE latch, TMR Two ways of protecting stored data against cosmic-ray upsets. Both cost chip area. Lockstep Running two processors on the same program and comparing them, so a fault is caught immediately. PMP Physical memory protection. Keeps parts of the software separate from each other. Cache A small fast memory. Usually faster, but occasionally much slower, which is why our processors do not have one. SiP System in package. Several chips inside one package. KGD Known-good die. Testing each chip before assembly, so one bad chip does not ruin a package. Screened Individually tested to a military standard. Screened chips sell for 10–100 times the commercial price. Sources: the DeepGrid investor brief and SKU architecture compendium; the defence ministry's 5th Positive Indigenisation List and DMA lists 1, 2, 3 and 5, read directly; UN COMTRADE HS 8542 for the import total; the generated DGridRiscV circuit code for §8; the ChipFoundry published shuttle schedule and price page and the EUROPRACTICE 2026 price list for §5 and §2, read on 4 August 2026. Volumes and prices are our own estimates unless a source is named. Performance curves in the drawings are example shapes against stated targets, not measurements.

LOI (letter of intent) — a signed letter from a customer saying they intend to buy. Our gates open on these.

Design-in — one customer engineering our chip into their product. Months of our support; five to ten per chip make its revenue.

Crash case — the version of a forecast where things go badly. Ours sits beside the plan everywhere the plan appears.

Stop rule — a decision written in advance — with a date — about what we do when a specific bad thing happens, so it cannot be renegotiated in an excited meeting.